HIGH-VOLTAGE ANALOG-SIGNAL IC

UC1672

4BPx120SEG Dot Matrix LCD Controller/Driver

ES Specifications
Datasheet Revision 0.6

IC Version c_B
June 13, 2023

ULTRACHIP

The Coolest LCD Driver, Ever!!

Table of Content

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4BPx120SEG Dot Matrix LCD Controller/Driver

INTRODUCTION

The UC1672 is a peripheral device which interfaces to almost any Liquid Crystal Display (LCD) with low multiplex rates. It generates the drive signals for any static or multiplexed LCD containing up to four (4) backplanes and up to eighty (120) SEGs and can easily be cascaded for larger LCD applications. The UC1672 is compatible with most microprocessors or microcontrollers and communicates via a two-line bidirectional I^2C-bus. Communication overheads are minimized by a display RAM with auto-increment addressing, by hardware sub-addressing, and by display memory switching (static and duplex drive modes).

MAIN APPLICATIONS

- Battery operated hand held devices
- Portable Instruments

FEATURE HIGHLIGHTS

- Single-chip LCD controller and driver
- Selectable backplane drive configuration: static, 2, 3, or 4 backplane multiplexing
- Selectable display bias configuration: static, $\frac{1}{2}$, or $\frac{1}{3}$
- Supports industry standard SP,I 3-wire(S9), 4-wire(S8) serial bus and I2C interface
- 120 SEG drives:
  - Up to 60 7-SEG alphanumeric characters
  - Up to 32 14-SEG alphanumeric characters
  - Any graphics of up to 480 elements
- 120 x 4 bit RAM for display data storage
- Supports both row-ordered and column-ordered display buffer RAM accesses
- .V_{DD} range (Typ.): 3V ~ 5.5V

  LCD V_{LCD} range: 2.5V ~ 5.5V(Step 0.5V)
- 400 kHz I^2C -bus interface
- May be cascaded for large LCD applications (up to 7680 SEGs possible)
- Available in gold bump die package.

Remark: The inspection standard of the product appearance is based on Ultrachip's inspection document.

ORDERING INFORMATION

Part Number	I^2C	Package	Description
UC1672cGAB	Yes	COG	Gold bumped die.

General Notes

APPLICATION INFORMATION
For improved readability, the specification contains many application data points. When application information is given, it is advisory and does not form part of the specification for the device.

USE OF I^2C
The implementation of I^2C is already included and tested in all silicon.

BARE DIE DISCLAIMER
All die are tested and are guaranteed to comply with all data sheet limits up to the point of wafer sawing. There is no post waffle saw/pack testing performed on individual die. Although the latest modern processes are utilized for wafer sawing and die pick-&-place into waffle pack carriers, UltraChip has no control of third party procedures in the handling, packing or assembly of the die. Accordingly, it is the responsibility of the customer to test and qualify their application in which the die is to be used. UltraChip assumes no liability for device functionality or performance of the die or systems after handling, packing or assembly of the die.

LIFE SUPPORT APPLICATIONS
These devices are not designed for use in life support appliances, or systems where malfunction of these products can reasonably be expected to result in personal injuries. Customer using or selling these products for use in such applications do so at their own risk.

CONTENT DISCLAIMER

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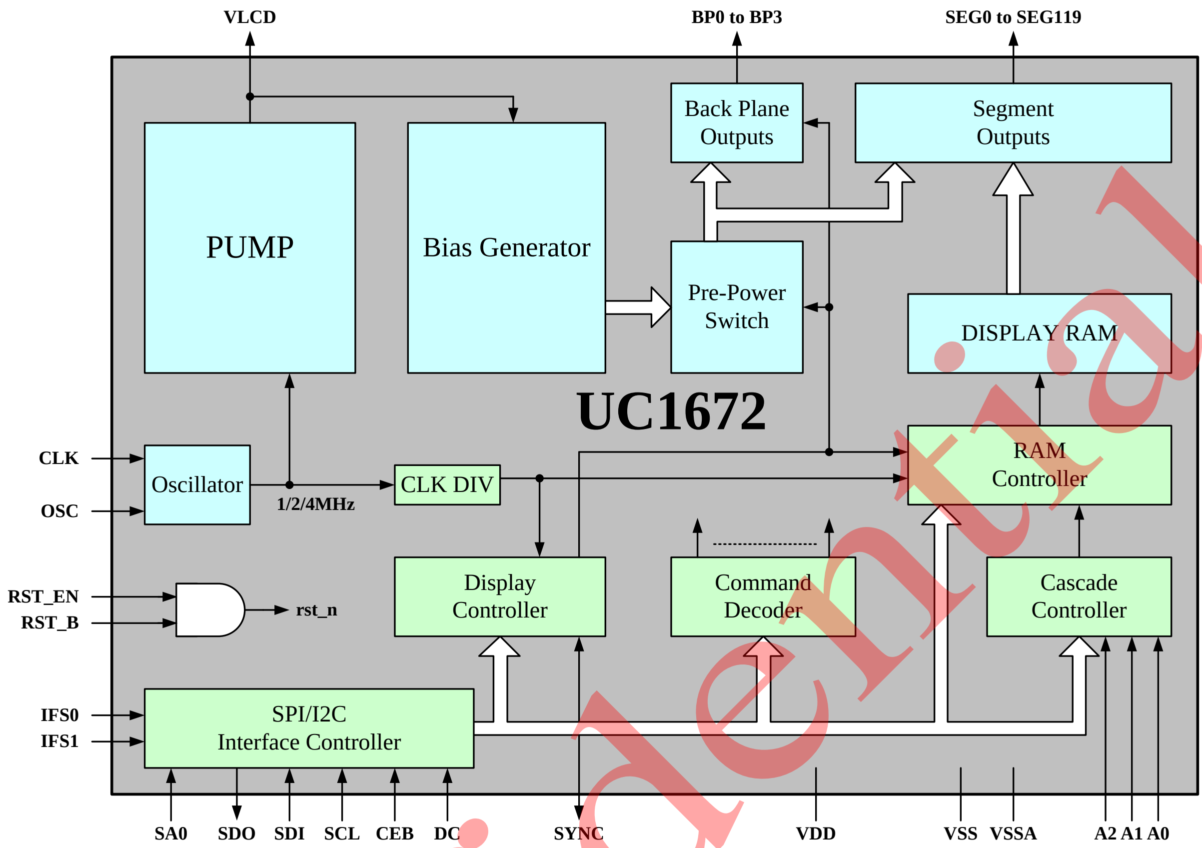
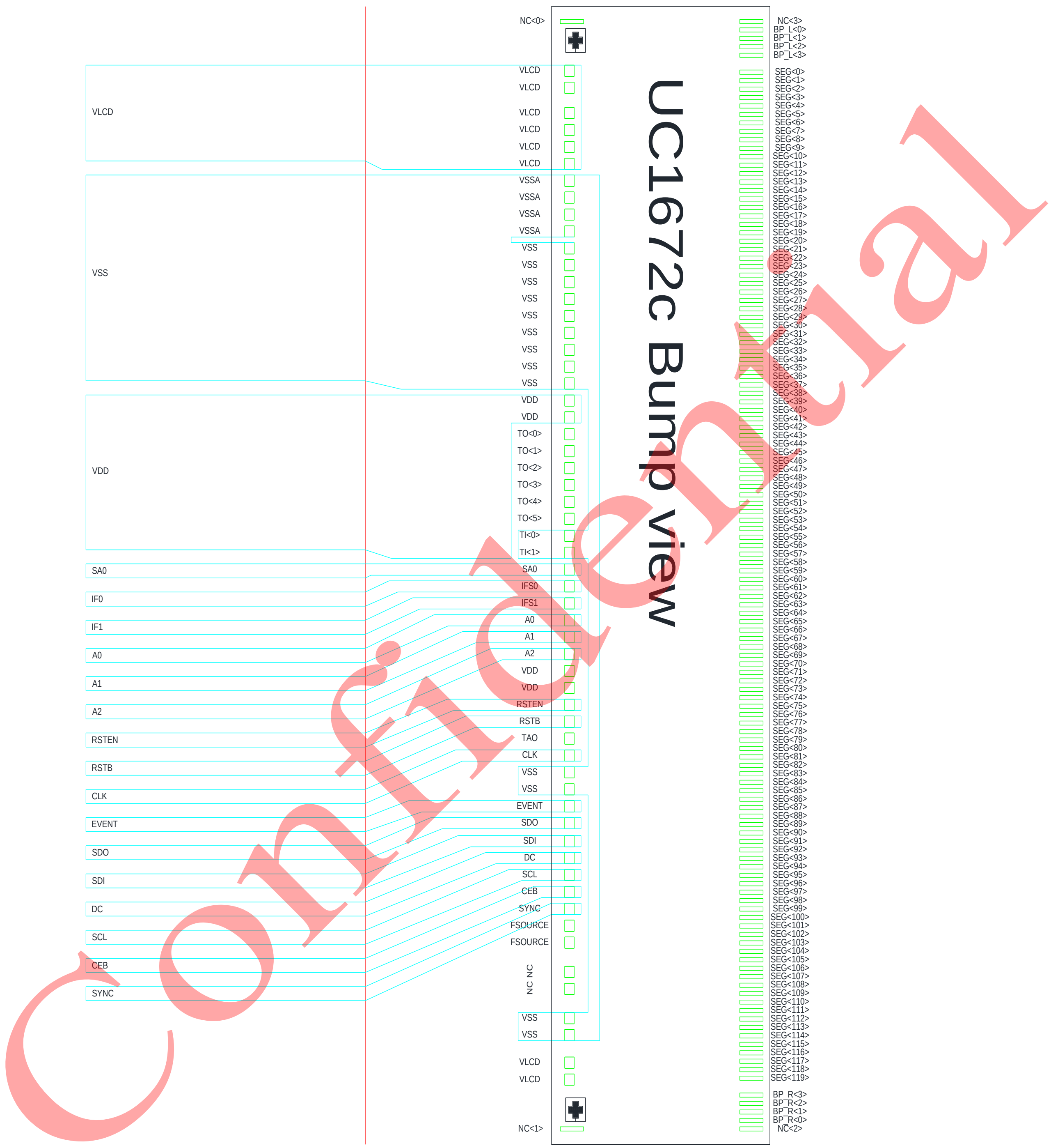
BLOCK DIAGRAM



PIN DESCRIPTION

<table>
<tr><th>Pin (Pad) Name</th><th>Pad No.</th><th>Type</th><th>Description</th></tr>
<tr><td colspan="4">Main Power Supply</td></tr>
<tr><td>V_{DD}</td><td>4</td><td>PWR</td><td>Power supply pin.</td></tr>
<tr><td>V_{SS}</td><td>13</td><td>GND</td><td>V_{SS} is Logic circuit ground.</td></tr>
<tr><td>V_{SSA}</td><td>4</td><td>GND</td><td>V_{SSA} is Analog circuit ground.</td></tr>
<tr><td colspan="4">LCD Power Supply & Voltage Control</td></tr>
<tr><td>VLCD</td><td>8</td><td>PWR</td><td>LCD supply voltage</td></tr>
<tr><td colspan="4">Host Interface</td></tr>
<tr><td>IF0
IF1</td><td>1
1</td><td>I</td><td>Bus mode. The interface bus mode is determined by IF[1:0] by the following relationship:
<table>
<tr><th>IF[1:0]</th><th>Mode</th></tr>
<tr><td>00</td><td>S8 SPI</td></tr>
<tr><td>01</td><td>S9 SPI</td></tr>
<tr><td>10</td><td>Standard SPI</td></tr>
<tr><td>11</td><td>I^2C</td></tr>
</table></td></tr>
<tr><td>RSTB</td><td>1</td><td>I</td><td>IC Reset Pin, Low Active.</td></tr>
<tr><td>RSTEN</td><td>1</td><td>I</td><td>0: RSTB disable
1: RSTB enable</td></tr>
<tr><td>CEB</td><td>1</td><td>I</td><td>SPI select signal. Connect to V_{DD} if not used.</td></tr>
<tr><td>DC</td><td>1</td><td>I</td><td>SPI DC signal. Connect to V_{DD} if not used.</td></tr>
<tr><td>SDO</td><td>1</td><td>O</td><td>I^2C /SPI data output</td></tr>
<tr><td>SDI</td><td>1</td><td>I</td><td>I^2C/SPI data input.</td></tr>
<tr><td>SCL</td><td>1</td><td>I</td><td>I^2C/SPI clock signal.</td></tr>
<tr><td>SA0</td><td>1</td><td>I</td><td>slave address selection
Connect to V_{SS} for logic 0, V_{DD} for logic 1.</td></tr>
<tr><td>A0, A1, A2</td><td>3</td><td>I</td><td>Hardware device address selection for cascading.</td></tr>
<tr><td>CLK</td><td>1</td><td>I/O</td><td>Clock input and output</td></tr>
<tr><td>$\overline{SYNC}$</td><td>1</td><td>I/O</td><td>Cascade synchronization input and output.</td></tr>
<tr><td>EVENT</td><td>1</td><td>O</td><td>Event triggers PIN. This PIN is triggered when there is an abnormal condition in the IC.</td></tr>
<tr><td>TI0, TI1</td><td>1,1</td><td>I</td><td>Test PIN, connect VSS if no use</td></tr>
<tr><td>TO0~TO5</td><td>6</td><td>O</td><td>Test PIN, leave these pins open during normal operation.</td></tr>
<tr><td>TAO</td><td>1</td><td>O</td><td>Test PIN, leave these pins open during normal operation.</td></tr>
<tr><td>FSOURCE</td><td>2</td><td>I</td><td>Test PIN, leave these pins open during normal operation.</td></tr>
<tr><td colspan="4">High Voltage LCD Driver Output</td></tr>
<tr><td>BP_R<0>, BP_L<0>
BP_R<1>, BP_L<1>
BP_R<2>, BP_L<2>
BP_R<3>, BP_L<3></td><td>1,1
1,1
1,1
1,1</td><td>HV</td><td>LCD backplane output</td></tr>
<tr><td>S0~S119
(SO<0>~SO<119>)</td><td>120</td><td>HV</td><td>LCD SEG output</td></tr>
<tr><td>NC
NC<0>
NC<1>
NC<2>
NC<3></td><td>2
1
1
1
1</td><td></td><td></td></tr>
</table>

RECOMMENDED COG LAYOUT



NOTES FOR V_{DD} WITH COG:

The operation condition, V_{DD}=3V (min.), should be satisfied under all operating conditions. UC1672c's peak current (I_{DD}) can be up to ~1mA during high speed data-write to UC1672c's on-chip SRAM. Such high pulsing current mandates very careful design of V_{DD} and V_{SS} ITO trances in COG modules. When V_{DD} and V_{SS} trace resistance is not low enough, the pulsing I_{DD} current can cause the actual on-chip V_{DD} to drop to below 3V and cause the IC to malfunction.

CONTROL REGISTERS

UC1672c contains registers, which control the chip operation. The following table is a summary of these control registers, a brief description and the default values. These registers can be modified by commands, which will be described in the next two sections, Command Table and Command Description.

Name: The Symbolic reference of the register.
Note that, some symbol name refers to bits (flags) within another register.

Default: Numbers shown in **Bold** font are default values after software reset and pin reset.

Name	Bits	Default	Description
DEV[2:0]	2	000b	3-bit binary value of 0 to 7, transferred to the sub-address counter to define one of 8 hardware sub-addresses.
BF[1:0]	2	00b	Blink frequency selection **00b: off** 01b: blink mode1 10b: blink mode 2 11b: blink mode 3
A	1	0b	Blink mode selection **0b: normal blinking.** **Normal blinking can only be selected when multiplex drive mode=1:3 or 1:4.** 1b: blinking by alternating display RAM banks
FR[2:0]	2	010b	SEG/COM Frame Rate Selection 000b: 60Hz 001b: 80Hz **010b: 100Hz** 011b: 120Hz 100b: 140Hz 101b: 160Hz 110b: 180Hz 111b: 200Hz
ES	1	0b	EVENT Pin setting **0b: Disable, EVENT PIN always output '0'.** 1b: Enable, when any one of BYTE_ERR/RST_DET is '1', then EVENT PIN will output '1'
CS[1:0]	2	00b	CLK Pin setting **00b: TCON will use internal OSC clock.** 01b: TCON will use internal OSC clock and also output this clock to CLK PIN. 10b: TCON will use external clock from CLK PIN. 11b: Reserved.
CPF[1:0]	2	11b	PUMP1 frequency divider setting 00b: 1MHz 10b: 2MHz 01b: 4MHz **11b: 8MHz**
VLCDMS[1:0]	2	00b	VLCD mode setting **00b: PUMP disabled; external supply of VLCD**. 01b: PUMP enabled, internal VLCD generation. 10b,11b: Direct mode enabled, VDD is internal connected to VLCD, PUMP disabled.
VSET[5:0]	6	32h	VLCD Voltage Setting (**default 5V**) VTFT = 2.5V + VSET * 0.05
APO	1	0b	All Pixel on control **0b: All pixel on disable** 1b: All pixel on enable
AXV	1	0b	All Pixel inverse **0b: Normal** 1b: Inverse
I	1	0b	Input bank selection, that is to select the storage of arriving display data **0b: RAM bit 0 (RAM bits 0 and 1)** 1b: RAM bit 2 (RAM bits 2 and 3)
O	1	0b	Output bank selection, that is to select the retrieval of LCD display data **0b: RAM bit 0 (RAM bits 0 and 1)** 1b: RAM bit 2 (RAM bits 2 and 3)

Name	Bits	Default	Description
M[1:0]	2	00b	LCD drive mode selection **00b: 1:4 multiplex; 4 backplanes** 01b: static; 1 backplane 10b: 1:2 multiplex; 2 backplanes 11b: 1:3 multiplex; 3 backplanes.
B	1	0b	LCD bias configuration **0b: 1/3 bias** 1b: 1/2 bias
E	1	0b	Display Enable **0b: display off** 1b: display on
DP[6:0]	7	00H	7-bit binary value of 0 to 119, transferred to the data pointer to define one of 120 display RAM addresses. Value range: 0000000b~1110111b, that is 0~119 in decimal
RAM_DAT[7:0]	8	00H	Display RAM Data
Status Registers			
ERR_SDOX	1	--	SDO Output conflict
ERR_CMD	1	--	Interface command error. 1b: Command write is not equal 0x50 ~ 0x5F. (Read clear to 0)
ERR_BYTE	1	--	The last bit of Interface transmission does not byte boundry. (Read will clear to 0.)
RST_DET	1	--	Reset Detection. If this is 1, then it means that PIN reset has occurred, and if it is 0, then it means no that PIN reset has occurred. Read will clean to 0.
CP_OK	1	--	VLCD Voltage is Ready

COMMAND TABLE

The following is a list of host commands supported by UC1672c
C/D: 0: Control, 1: Data D7-D0: #: Useful Data bits –: Don't Care,.
When DA2/DA1/DA0 matches the PIN A2/A1/A0 of the IC, data will be written, otherwise data will be ignored. Except for code 0x0 and code 0x2, all devices will be executed at the same time.

#	Command	C/D	D7	D6	D5	D4	D3	D2	D1	D0	Action	Default
1	Software Reset	0	0	1	0	1	0	0	0	0	Software Reset	50H
2	Select Device	0	0	1	0	1	0	0	0	1	DEV[2:0]	51H
		1	0	0	0	0	0	#	#	#		00H
3	Select Frame-Rate & Blink	0	0	1	0	1	0	0	1	0	BF[1:0],A,FR[2:0]	52H
		1	0	0	#	#	#	#	#	#		02H
4	Pump Control	0	0	1	0	1	0	0	1	1	ES,CS[1:0],CPF[1:0], VLDMS[1:0]	53H
		1	0	#	#	#	#	#	#	#		0CH
5	Pump Voltage Set	0	0	1	0	1	0	1	0	0	VSET[5:0]	54H
		1	0	0	#	#	#	#	#	#		32H
6	Display Mode	0	0	1	0	1	0	1	0	1	APO,AXV,I.O,M[1:0] B,E	55H
		1	#	#	#	#	#	#	#	#		00H
7	Select RAM Address/Data	0	0	1	0	1	0	1	1	0	Select RAM Address	56H
		1	0	#	#	#	#	#	#	#		00H
		1	#	#	#	#	#	#	#	#	RAM_DAT[7:0]	00H
8	Get Status	0	0	1	0	1	1	1	1	0	Get Status	5EH
		1	-	-	#	#	#	#	-	#		N/A

COMMAND DESCRIPTION

(1) SOFTWARE RESET

Action	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Software Reset	0	0	1	0	1	0	0	0	0

This command is Software reset; all register will reset to default value. Please wait for at least 3mS after Software Reset before issuing any commands. When the “Software Reset” command is issued, it is received and executed by all the ICs (including the ICs connected in cascade).

(2) SELECT DEVICE

Action	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Select Device	0	0	1	0	1	0	0	0	1
	1	0	0	0	0	0	DEV[2:0]		

DEV[2:0]: 3-bit binary value of 0 to 7, transferred to the sub-address counter to define one of 8 hardware sub-addresses.

When the “Select Device” command is issued, it is received and executed by all the ICs (including the ICs connected in cascade).

(3) SELECT FRAME-RATE & BLINK

Action	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Select Frame rate & Blink	0	0	1	0	1	0	0	1	0
	1	0	0	BF[1:0]		A	FR[2:0]		

BF[1:0]: blink frequency selection

00b: off 01b: blink mode1 10b: blink mode 2 11b: blink mode 3

See the Blink Frequency table for detailed description about each blink mode.

A: blink mode selection

0b: normal blinking. Normal blinking can only be selected when multiplex drive mode=1:3 or 1:4.
1b: blinking by alternating display RAM banks

FR[2:0]: SEG/COM Frame Rate Selection

000b: 60Hz 100b: 140Hz
001b: 80Hz 101b: 160Hz
010b: 100Hz 110b: 180Hz
011b: 120Hz 111b: 200Hz

When the “Select Frame rate & Blink” command is issued, it is received and executed by all the ICs (including the ICs connected in cascade).

(4) PUMP CONTROL

Action	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Pump Control	0	0	1	0	1	0	0	1	1
	1	0	ES	CS[1:0]		CPF[1:0]		VLCDMS[1:0]	

ES: EVENT Pin setting

0b: Disable, EVENT PIN always output '0'.

1b: Enable, when any one of BYTE_ERR/RST_DET is '1', then EVENT PIN will output '1'

CS[1:0]: CLK Pin setting

00b: TCON will use internal OSC clock.

01b: TCON will use internal OSC clock and also output this clock to CLK PIN.

10b: TCON will use external clock from CLK PIN.

11b: Reserved.

CPF[1:0]: PUMP1 frequency divider setting

00b: 1MHz 10b: 2MHz 01b: 4MHz **11b: 8MHz**

VLCDMS[1:0]: VLCD mode setting

00b: PUMP disabled; external supply of VLCD.
01b: PUMP enabled, internal VLCD generation.
10b,11b: Direct mode enabled, VDD is internal connected to VLCD, PUMP disabled.

(5) PUMP VOLTAGE SET

Action	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Pump Voltage Set	0	0	1	0	1	0	1	0	0
	1	0	0	VSET[5:0]					

VSET[5:0]: VLCD Voltage Setting, VLCD = 2.5 + VSET * 0.05(**default 5V**).

When the “Pump Voltage Set” command is issued, it is received and executed by all the ICs (including the ICs connected in cascade).

(6) DISPLAY MODE

Action	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Display Mode	0	0	1	0	1	0	1	0	1
	1	APO	AXV	I	O	M[1:0]		B	E

When the “Display Mode” command is issued, it is received and executed by all the ICs (including the ICs connected in cascade).

APO: All Pixel On control

0b: All pixel on disable

1b: All pixel on enable

AXV: All Pixel On Inverse

0b: Normal

1b: Inverse

I: Input bank selection, that is to select the storage of arriving display data

0b: RAM bit 0 (RAM bits 0 and 1)

1b: RAM bit 2 (RAM bits 2 and 3)

O: Output bank selection, that is to select the retrieval of LCD display data

0b: RAM bit 0 (RAM bits 0 and 1)

1b: RAM bit 2 (RAM bits 2 and 3)

M[1:0]: LCD drive mode selection

00b: 1:4 multiplex; 4 backplanes

01b: static; 1 backplane

10b: 1:2 multiplex; 2 backplanes

11b: 1:3 multiplex; 3 backplanes

B: LCD bias configuration

0b: 1/3 bias

1b: 1/2 bias

E: Display Enable

0b: display off

1b: display on

(7) SELECT RAM ADDRESS/DATA

Action	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Select RAM Address/Data	0	0	1	0	1	0	1	1	0
	1	0	DP[6:0]						
	1	RAM_DAT[7:0]							
	1	……							
	1	RAM_DAT[7:0]							

DP[6:0]: 7-bit binary value of 0 to 119, transferred to the data pointer to define one of 120 display RAM addresses.

Value range: 0000000b~1110111b, that is 0~119 in decimal**(default 00H)**

RAM_DAT[7:0]: Display RAM data

(8) GET STATUS

Action	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Get Status	0	0	1	0	1	1	1	1	0
	1	-	-	ERR_SDOX	ERR_CMD	ERR_BYTE	RST_DET	-	CP_OK

ERR_SDOX: SDO Output conflict

ERR_CMD: Interface command error.

1b: Command write is not equal 0x50 ~ 0x5F. (Read clear to 0)

ERR_BYTE: The last bit of interface transmission does not byte boundary. (Read will clear to 0)

RST_DET: Reset Detection. If this is 1, then it means that PIN reset has occurred,

and if it is 0, then it means that no PIN reset has occurred. Read will clean to 0.

CP_OK: VLCD Voltage is Ready.

HOST INTERFACE

As summarized in the table below, UC1672c supports two serial bus protocols. Designers can choose to achieve high data transfer rate, or use serial bus to create compact LCD modules and minimize connector pins.

		Bus Type			
		S8 (4-wire)	Standard SPI	S9 (3-wire)	I2C
Width		Serial			
Control & Data Pins	IF[1:0]	00	10	01	11
	CEB	Chip select			-
	DC	Control /Data	-		
	SCL	clock signal.			
	SDI	data input signal.			
	SDO	data output signal.			

* Connect unused control pins and data bus pins to V_{DD} or V_{SS}.

	CEB Disable Bus Interface	CEB Init. Bus State	RESET Init. Bus State
I2C	–	–	✓
S9	✓	✓	✓
S8	✓	✓	✓
Standard	✓	✓	✓

- CEB disable bus interface – CEB can be used to disable Bus Interface Write / Read Access.
- RESET can be pin reset.

Table 3: Host interfaces Summary

S8 (4-wire) Interface

Pins CEB are used for chip select and bus cycle reset. Pin DC is used to determine the content of the data been transferred.
During each write cycle, 8 bits of data, MSB first, are latched on eight rising SCL edges into an 8-bit data holder.
If DC=0, the data byte will be decoded as command. If DC=1, this 8-bit will be treated as data and transferred to proper address in the Display Data RAM on the rising edge of the last SCL pulse.

S8-bus slave address

Device selection depends on the S8-bus slave address.

S8 slave address byte

7	6	5	4	3	2	1	0
1	0	1	0	1	0	SA0	R /W

R/ $\overline{W}$: 0b: write data   1b: read data

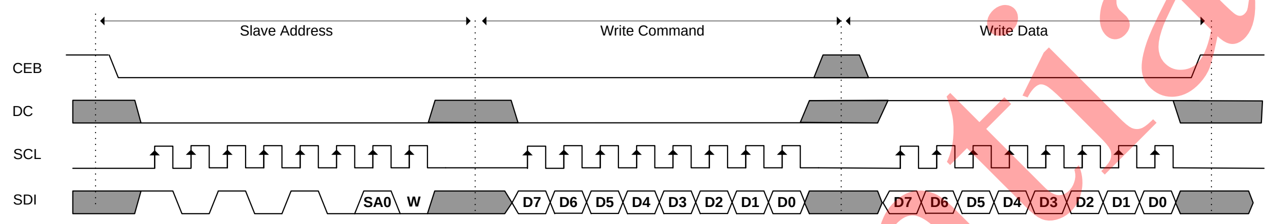


FIGURE 4.a: 4-wire Serial Interface (S8) – Write

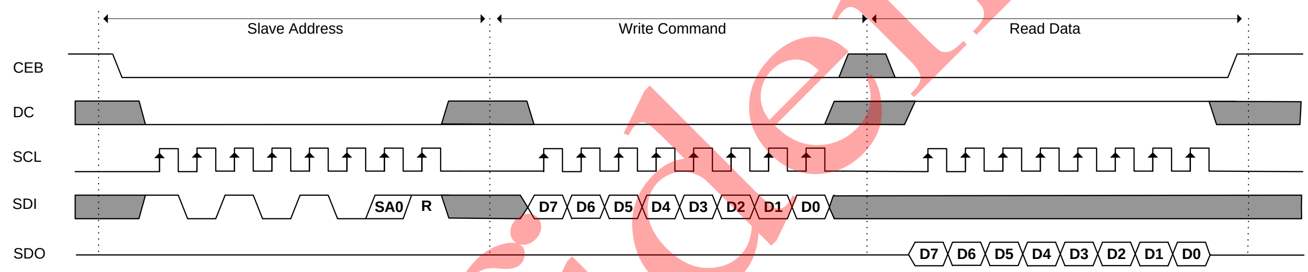


Figure 4.b: 4-wire Serial Interface (S8) – Read

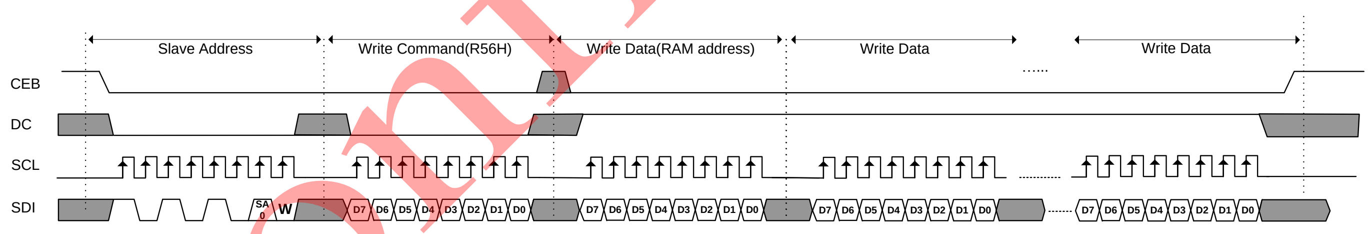


FIGURE 4.c: 4-wire Serial Interface (S8) – Write SRAM

S9 (3-wire) Interface

The Serial Peripheral Interface (SPI) bus provides access to a flexible, full-duplex synchronous serial bus. SPI can operate as a slave device in 3-wire modes, and supports multiple slaves on a single SPI bus. The chip-select (CEB) signal is an input to select SPI in slave mode. Data on SDI are sampled in the center of each data bit.

S9-bus slave address

Device selection depends on the S9-bus slave address.

S9 slave address byte

7	6	5	4	3	2	1	0
1	0	1	0	1	0	SA0	R/W

R/$\overline{W}$: 0b: write data 1b: read data

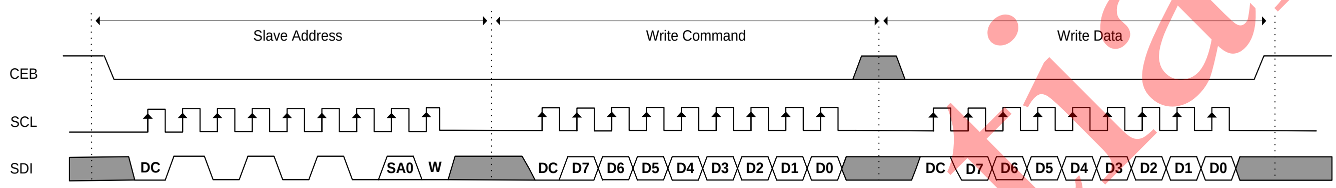


FIGURE 5.a: 3-wire Serial Interface (S9) – Write

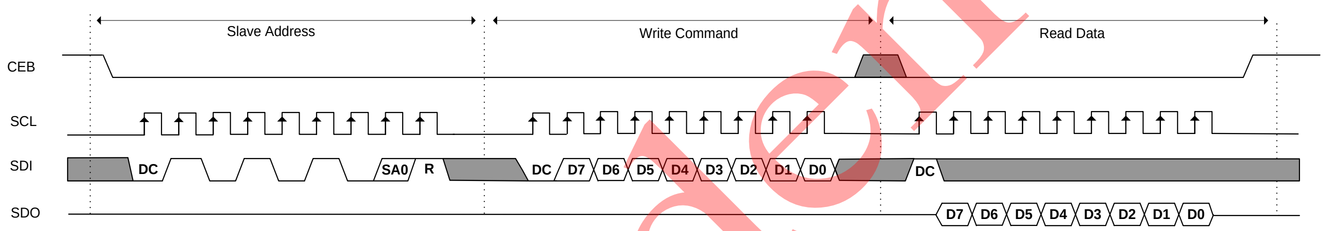


FIGURE 5.b: 3-wire Serial Interface (S9) – Read

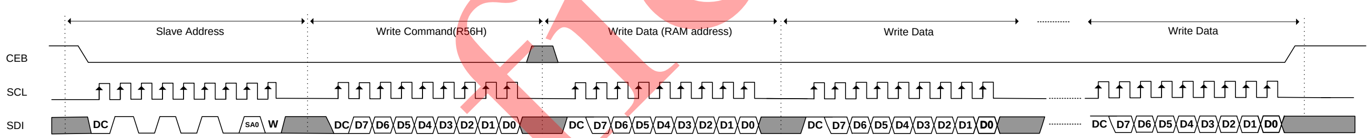


FIGURE 5.c: 3-wire Serial Interface (S9) – Write SRAM

Standard SPI Interface

The Serial Peripheral Interface (SPI) bus provides access to a flexible, full-duplex synchronous serial bus. SPI can operate as a slave device in 4-wire modes, and supports multiple slaves on a single SPI bus. The chip-select (CSB) signal is an input to select SPI in slave mode. Data on DIN are sampled in the center of each data bit. See section "AC Characteristics" for timing parameters.

Signal Description of Standard SPI-bus

CEB (Chip Select) The CSB signal is an input to slave devices. CEB=0 selects the SPI device.
SCL (Data Clock) The SCL signal is an input to slave devices. It is used to synchronize the transfer of data between the master and slave. The SCL signal is ignored by a SPI slave when the slave is not selected (CEB=1).
SDI(Data In) The DIN signal is an input to slave devices. It is used to serially transfer data from the master to the slave. Data is transferred most-significant bit first.
SDO(Data Out) The DOUT signal is an output from a slave device. It is used to serially transfer data from the slave to the master. Data are transferred most-significant bit first.

Standard SPI-bus slave address

Device selection depends on the S8-bus slave address.

Standard SPI slave address byte

7	6	5	4	3	2	1	0
1	0	1	0	1	0	SA0	R/$\overline{W}$

R/$\overline{W}$: 0b: write data 1b: read data

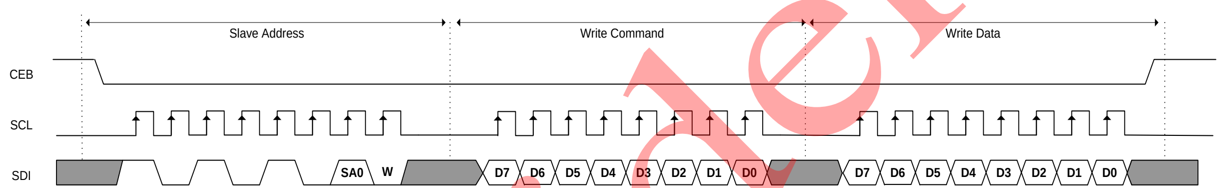


FIGURE 6.a: Standard Serial Interface – Write

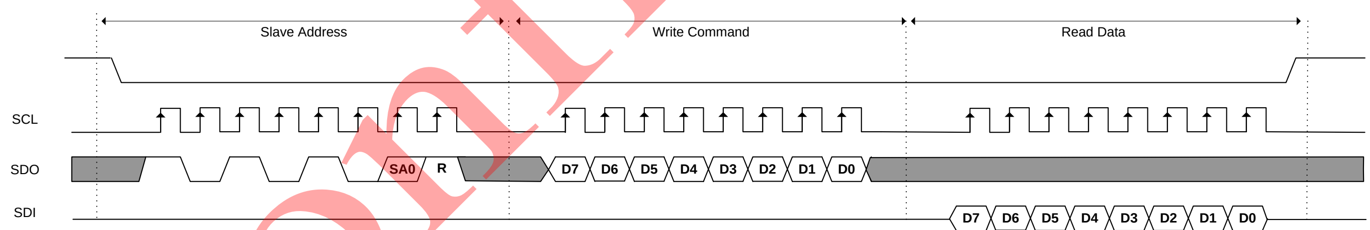


FIGURE 6.b: Standard Serial Interface – Read

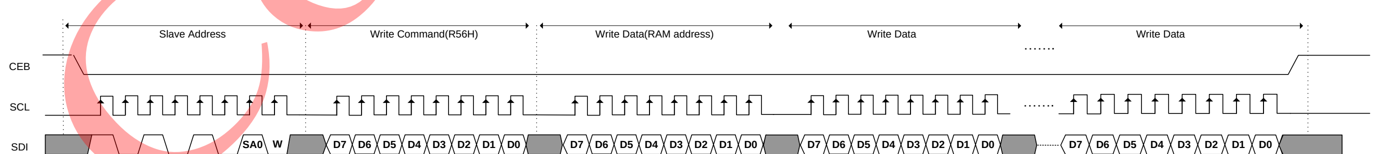


FIGURE 6.c: Standard Serial Interface – Write SRAM

I2C interface

The I2C-bus is for bidirectional, two-line communication between different ICs or modules. The two lines are a Serial Data line (SDI) and a Serial CLock line (SCL). Both lines must be connected to a positive supply via a pull-up resistor when connected to the output stages of a device. Data transfer may be initiated only when the bus is not busy.

In Chip-On-Glass (COG) applications, where the track resistance between the SDO output pin to the system SDI input line can be significant, the bus pull-up resistor and the Indium Tin Oxide (ITO) track resistance may generate a voltage divider. As a consequence it may be possible that the acknowledge cycle, generated by the LCD driver, cannot be interpreted as logic 0 by the master. Therefore it is an advantage for COG applications to have the acknowledge output separated from the data line. For that reason, the SDI line of the UC1672 is split into SDI and SDO.

In COG applications where the acknowledge cycle is required, it is necessary to minimize the track resistance from the SDO pin to the system SDI line to guarantee a valid LOW level.

By splitting the SDI line into SDI and SDO (having the SDO open circuit), the device could be used in a mode that ignores the acknowledge cycle. Separating the acknowledge output from the serial data line can avoid design efforts to generate a valid acknowledge level. However, in that case the I2C-bus master has to be set up in such a way that it ignores the acknowledge cycle.2

By connecting pin SDO to pin SDI the SDI line becomes fully I2C-bus compatible. The following definition assumes SDI and SDO are connected and refers to the pair as SDI

I^2C-bus slave address

Device selection depends on the I2C-bus slave address.

I^2C slave address byte

7	6	5	4	3	2	1	0
1	0	1	0	1	0	SA0	R /$\overline{W}$

R/ $\overline{W}$: 0b: write data 1b: read data

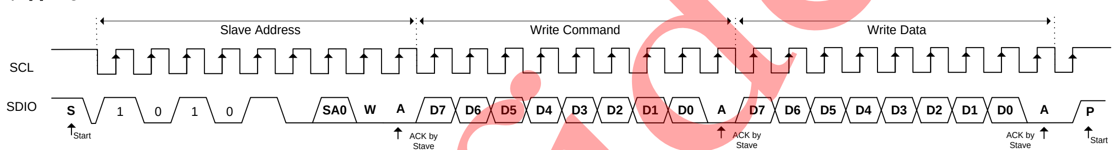


FIGURE 7a: I2C Interface – Write

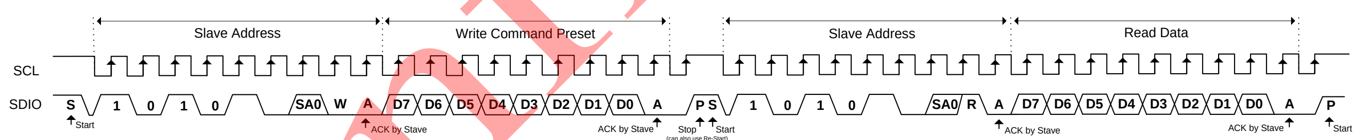


FIGURE 7b: I2C Interface – Read

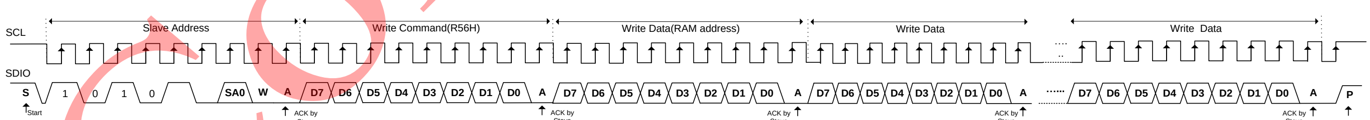


FIGURE 7c: I2C Interface – Write SRAM

Bit transfer

One data bit is transferred during each clock pulse. The data on the SDI line must remain stable during the HIGH period of the clock pulse as changes in the data line at this time are interpreted as a control signal (see Figure 25).

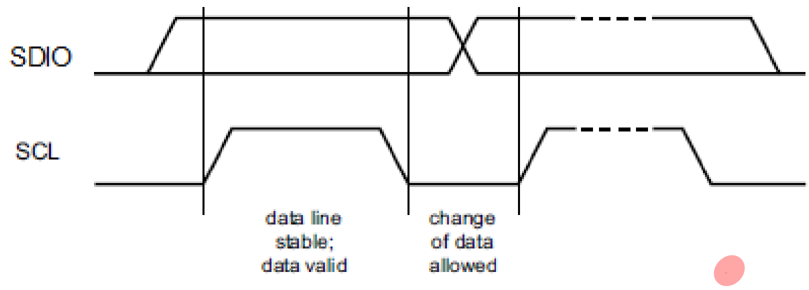


Figure 25. I^2C-bus - bit transfer

START and STOP conditions

Both data and clock lines remain HIGH when the bus is not busy.

A HIGH-to-LOW change of the data line, while the clock is HIGH is defined as the START condition (S).

A LOW-to-HIGH change of the data line while the clock is HIGH is defined as the STOP condition (P).

The START and STOP conditions are shown in Figure 26.

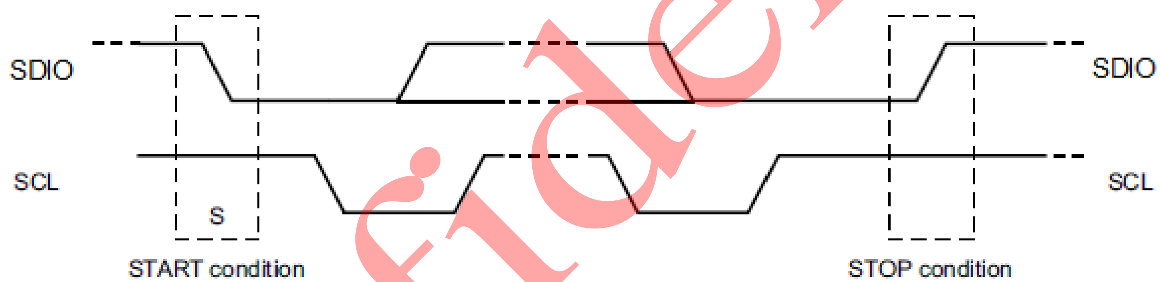


Figure 26. I^2C-bus - definition of START and STOP conditions

Acknowledge

The number of data bytes transferred between the START and STOP conditions from transmitter to receiver is unlimited. Each byte of 8 bits is followed by an acknowledge cycle.

- A slave receiver which is addressed must generate an acknowledge after the reception of each byte.
- Also a master receiver must generate an acknowledge after the reception of each byte that has been clocked out of the slave transmitter.
- The device that acknowledges must pull-down the SDIO line during the acknowledge clock pulse, so that the SDIO line is stable LOW during the HIGH period of the acknowledge related clock pulse (set-up and hold times must be considered).
- A master receiver must signal an end of data to the transmitter by not generating an acknowledge on the last byte that has been clocked out of the slave. In this event, the transmitter must leave the data line HIGH to enable the master to generate a STOP condition.

Acknowledgement on the I2C-bus is shown in Figure 28.

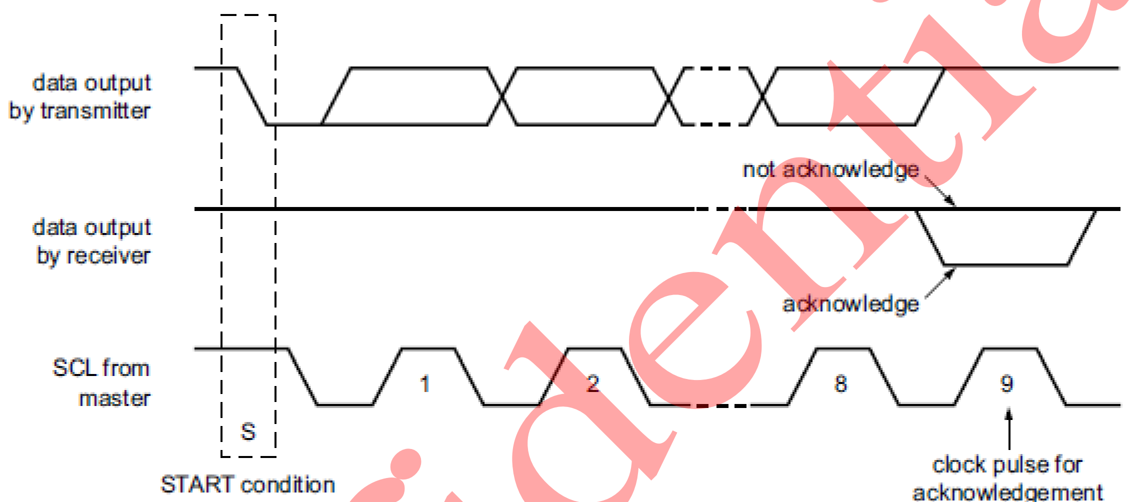
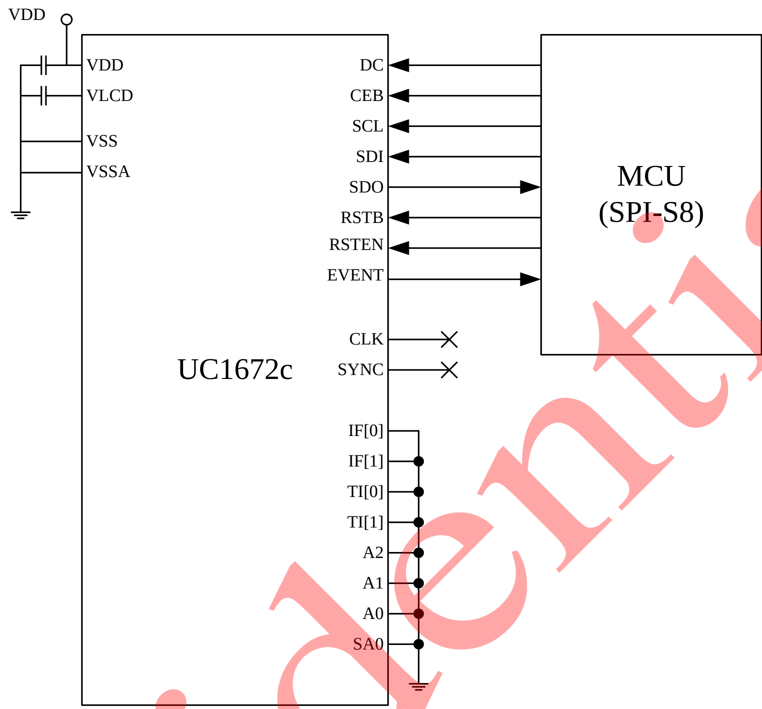


Figure 28. Acknowledgement on the I2C-bus

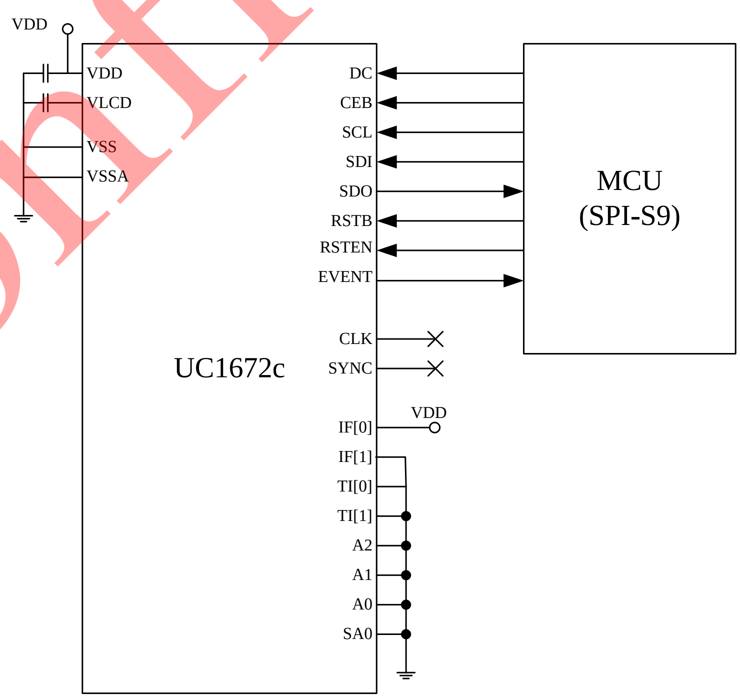
APPLICATION CIRUIT

1. Single chip cascade

- S8 SPI interface

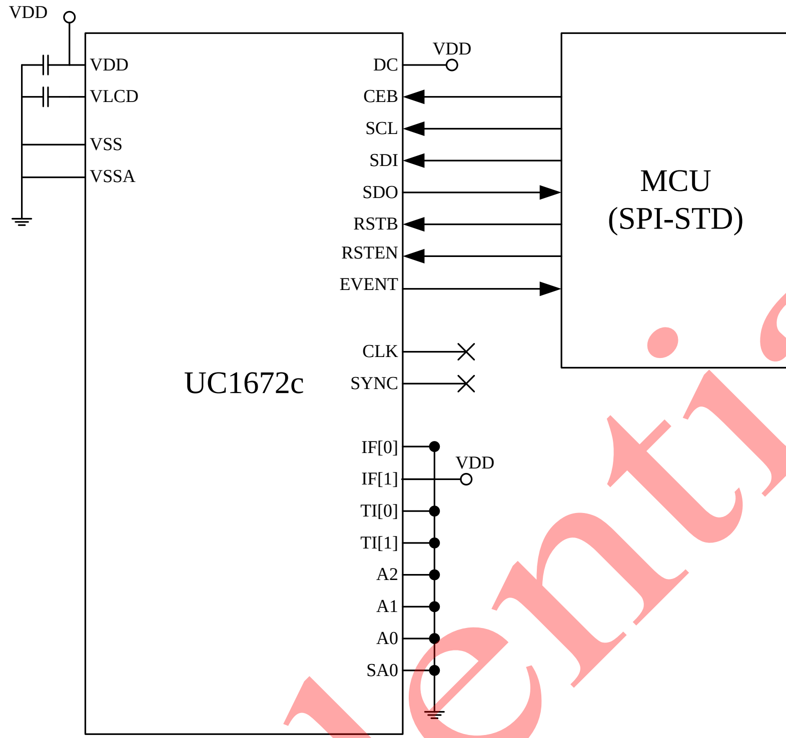


- S9 SPI interface

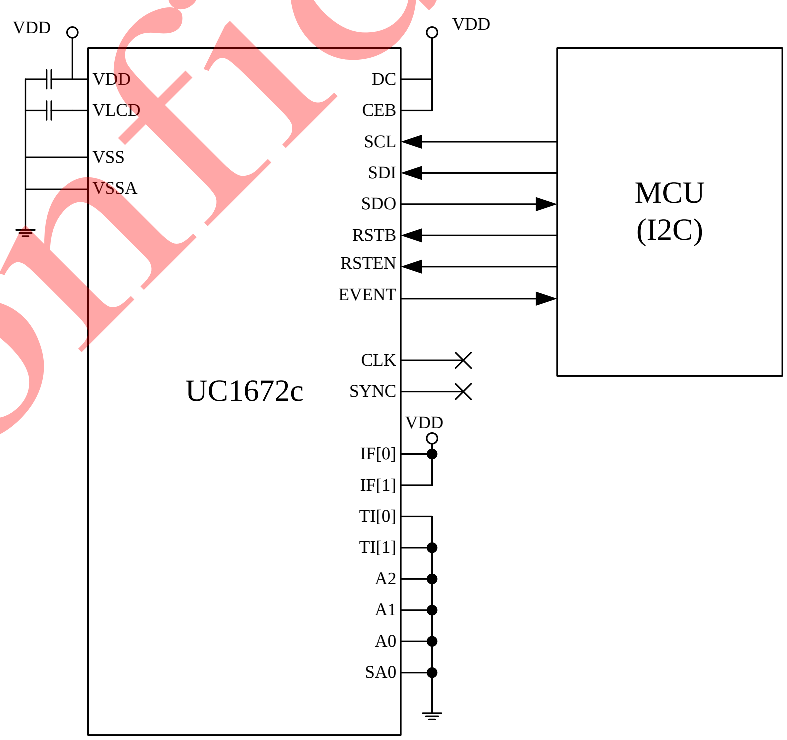


Note1: All CAP use 1uF

- Standard SPI interface



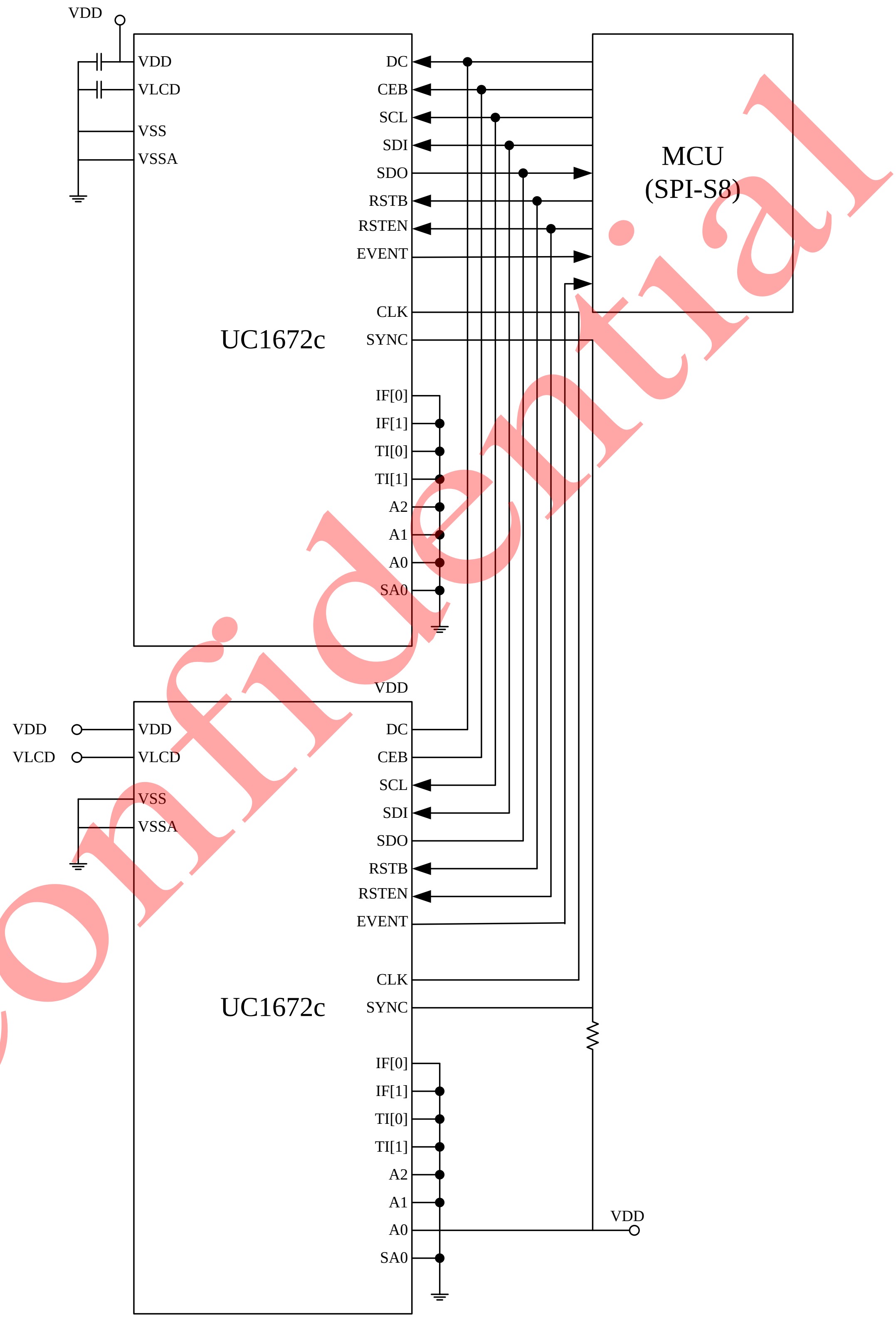
- I2C interface



Note: All CAP use 1uF

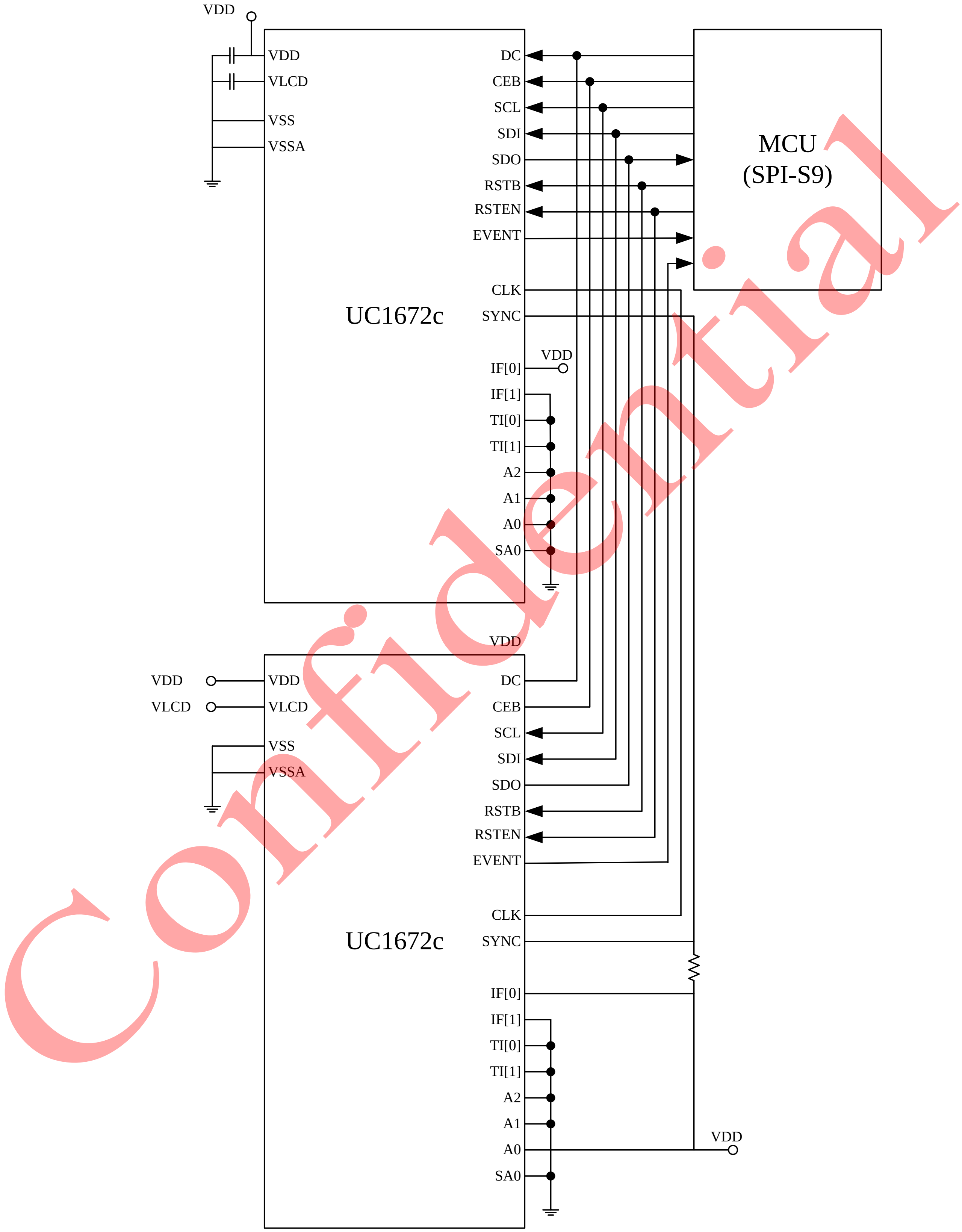
2. Two chip cascade

- S8 SPI interface



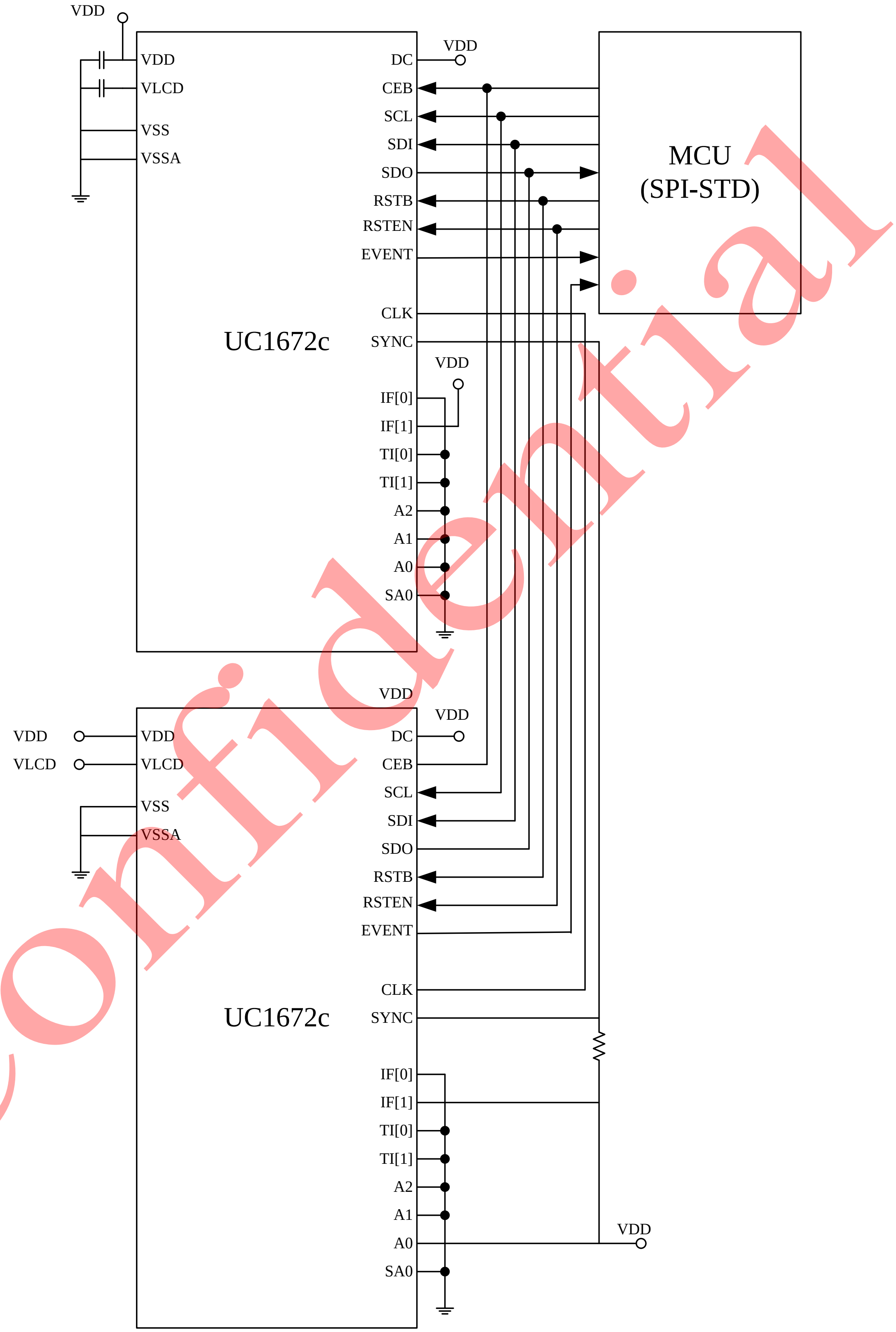
Note: All CAP use 1uF

- S9 SPI interface



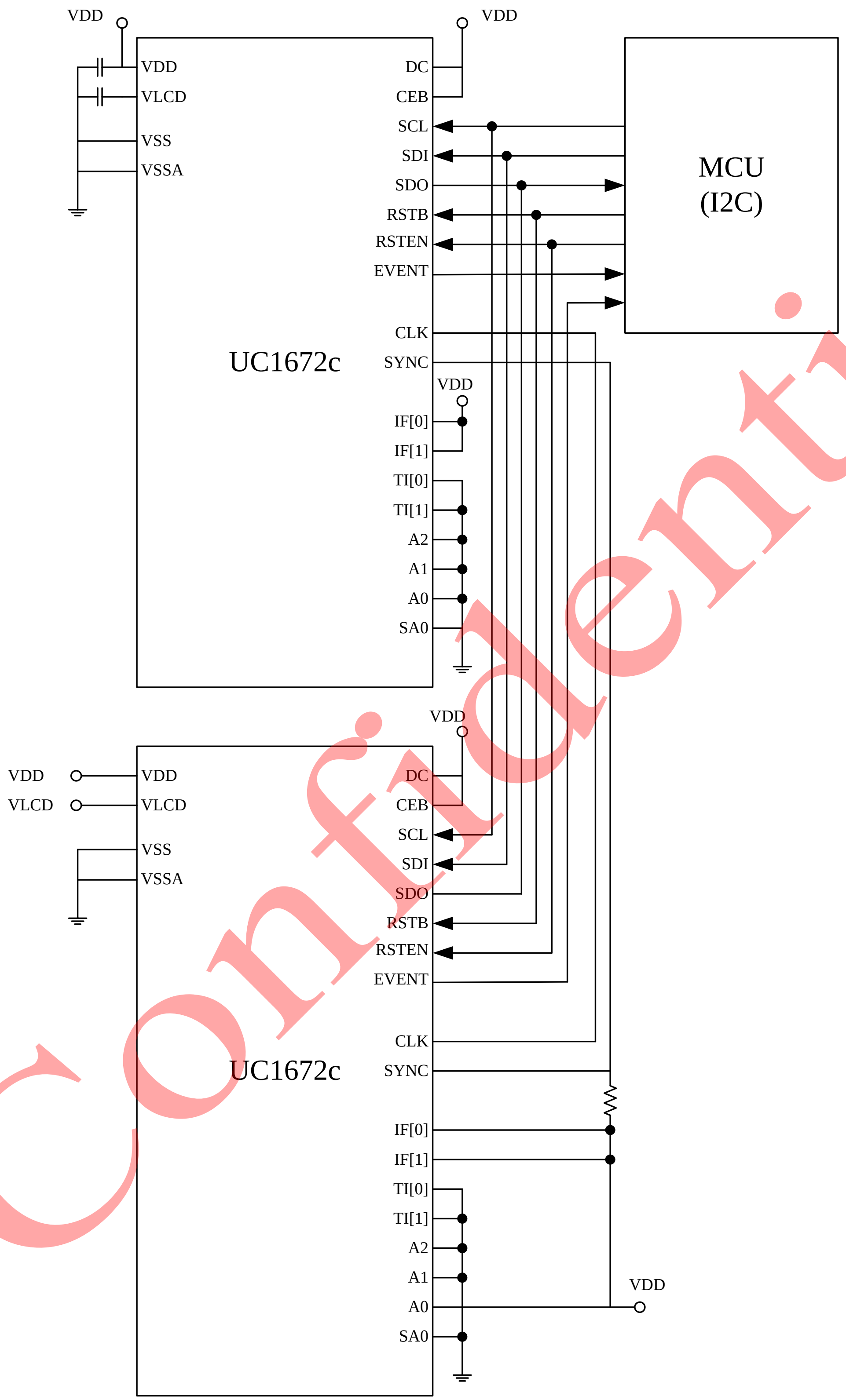
Note: All CAP use 1uF

- Standard SPI interface



Note: All CAP use 1uF

- I2C interface



Note: All CAP use 1uF

BLOCK FUNCTION DESCRIPTION

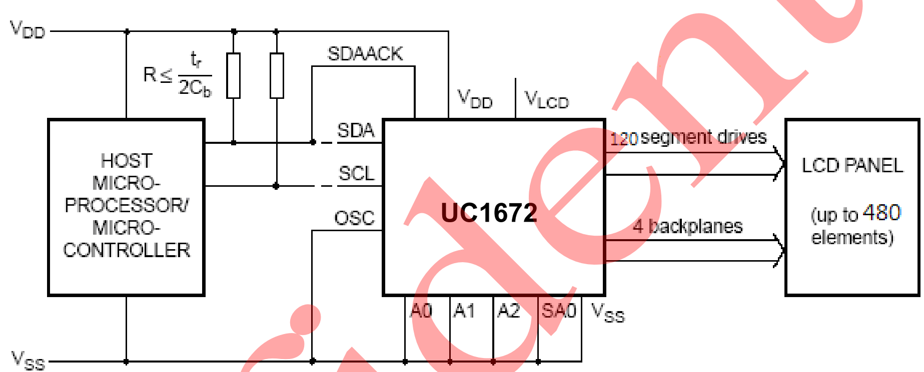
The UC1672 is a versatile peripheral device, designed to interface between any microprocessor or microcontroller to a wide variety of LCDs. It can directly drive any static or multiplexed LCD containing up to four backplanes and up to 120 SEGs.

The display configurations possible with the UC1672 depend on the required number of active backplane outputs. A selection of display configurations is shown in table below:

Possible Display Configurations

# of Backplanes	# of Elements	7-SEG alphanumeric		14-SEG alphanumeric		Dot matrix
		Digits	Indicator Symbols	Characters	Indicator Symbols	
4	480	60	60	32	32	480 (4x120)
3	360	45	45	24	24	360 (3x120)
2	240	30	30	16	16	240 (2x120)
1	120	15	15	8	8	120 (1x120)

All of the display configurations can be implemented in a typical system as shown in following figure:



The host microprocessor or microcontroller maintains the 2-line I²C-bus communication channel with the UC1672.

The internal oscillator is selected by connecting pin OSC to Vss. The only other connections required to complete the system are the power supplies (VDD, Vss, and VLCD) and the LCD panel selected for the application.

LCD Bias Generator

Fractional LCD biasing voltages are obtained from an internal voltage divider of three series resistors between VLCD and Vss. The center resistor is bypassed by switch if the $\frac{1}{2}$ bias voltage level for the 1:2 multiplex configuration is selected.

LCD Voltage Selector

The LCD voltage selector coordinates the multiplexing of the LCD in accordance with the selected LCD drive configuration. The operation of the voltage selector is controlled by the `Mode-Set` command in the `Command Description` section. The biasing configurations that apply to the preferred modes of operation, together with the biasing characteristics as functions of V_{LCD} and the resulting discrimination ratios (D), are given in table below:

LCD Drive Mode	# of Backplanes	# of Levels	LCD Bias Configuration	$\frac{V_{OFF(RMS)}}{V_{LCD}}$	$\frac{V_{ON(RMS)}}{V_{LCD}}$	$D=\frac{V_{ON(RMS)}}{V_{OFF(RMS)}}$
static	1	2	static	0	1	∞
1:2 multiplex	2	3	1/2	0.354	0.791	2.236
1:2 multiplex	2	4	1/3	0.333	0.745	2.236
1:3 multiplex	3	4	1/3	0.333	0.638	1.915
1:4 multiplex	4	4	1/3	0.333	0.577	1.732

A practical value for V_{LCD} is determined by equating $V_{OFF(RMS)}$ with a defined LCD threshold voltage (Vth), typically when the LCD exhibits approximately 10% contrast. In the static drive mode a suitable choice is V_{LCD} > 3Vth.

Multiplex drive modes of 1:3 and 1:4 with $\frac{1}{2}$ bias are possible but the discrimination and hence the contrast ratios are smaller.

Bias is calculated by $\frac{1}{1+a}$,

where a = 1 for $\frac{1}{2}$ bias

a = 2 for $\frac{1}{3}$ bias

The RMS on-state voltage ($V_{ON(RMS)}$) for the LCD is calculated with Equation (1):

$$V_{ON(RMS)} = V_{LCD}\sqrt{\frac{a^2+2a+n}{n\,x(1+a)^2}} \quad \text{.......... (1)}$$

where n = 1 for static mode
n = 2 for 1:2 multiplex
n = 3 for 1:3 multiplex
n = 4 for 1:4 multiplex

The RMS off-state voltage ($V_{OFF(RMS)}$) for the LCD is calculated with Equation (2):

$$V_{OFF(RMS)} = V_{LCD}\sqrt{\frac{a^2-2a+n}{n\,x(1+a)^2}} \quad \text{.......... (2)}$$

Discrimination is the ratio of $V_{ON(RMS)}$ to $V_{OFF(RMS)}$ and is determined from Equation (3):

$$D = \frac{Von(rms)}{Voff(rms)} = \sqrt{\frac{(a+1)^2+(n-1)}{(a-1)^2+(n-1)}} \quad \text{........ (3)}$$

where the discrimination for an LCD drive mode of 1:3 multiplex with $\frac{1}{2}$ bias is $\sqrt{3}$ =1.732 and

the discrimination for an LCD drive mode of 1:4 multiplex with $\frac{1}{2}$ bias is $\frac{\sqrt{21}}{3}$ =1.528.

The advantage of these LCD drive modes is a reduction of the LCD full scale voltage V_{LCD}, shown as follows:

- 1:3 multiplex ($\frac{1}{2}$ bias):

$$V_{LCD} = \sqrt{6} \text{ x } V_{OFF(RMS)} = 2.449\ V_{OFF(RMS)}$$

- 1:4 multiplex ($\frac{1}{2}$ bias):

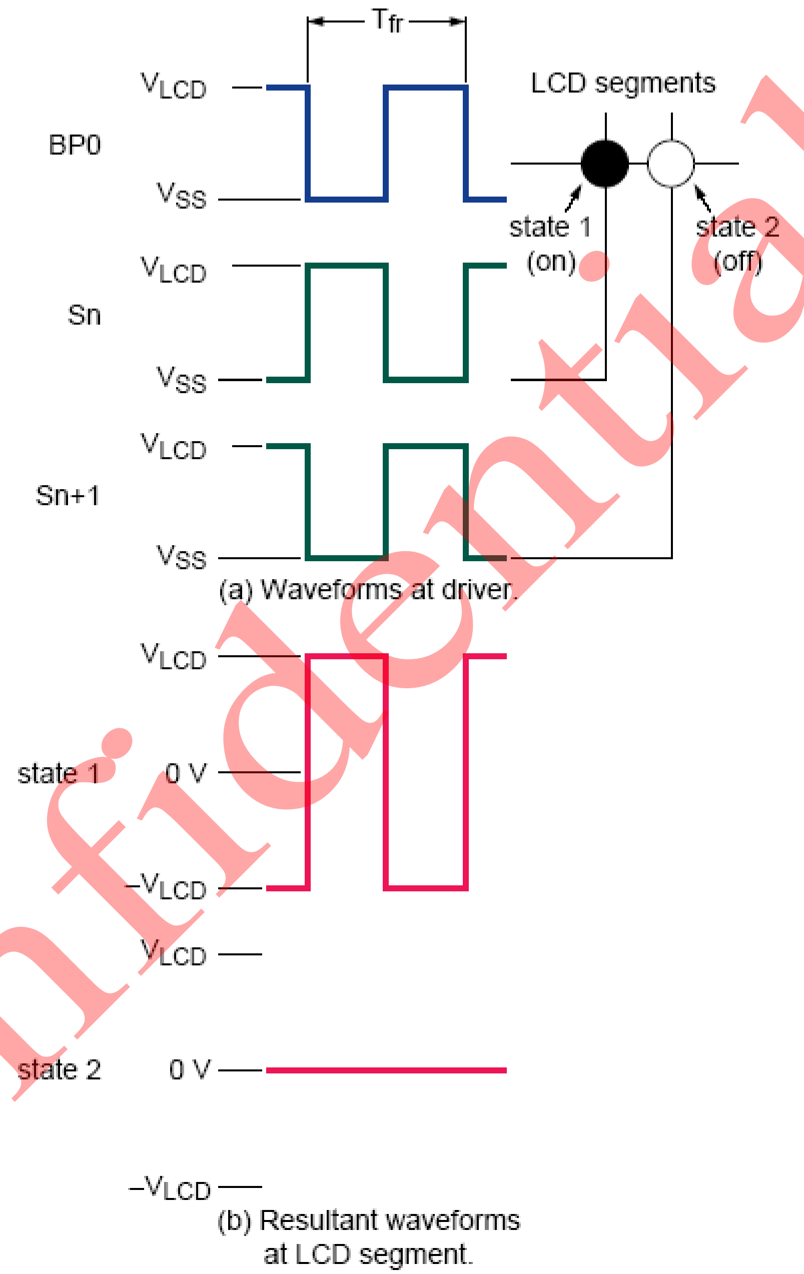
$$V_{LCD} = [\frac{(4x\sqrt{3})}{3}] = 2.309\ V_{OFF(RMS)}$$

These compare with V_{LCD}=3 $V_{OFF(RMS)}$ when $\frac{1}{3}$ bias is used.

It should be noted that V_{LCD} is sometimes referred as the LCD operating voltage.

LCD Drive Mode Waveforms – Static drive mode

The static LCD drive mode is used when a single backplane is provided in the LCD. Backplane and SEG drive waveforms for this mode are shown in the figure below:



$V_{state1}(t) = V_{Sn}(t) - V_{BP0}(t)$.

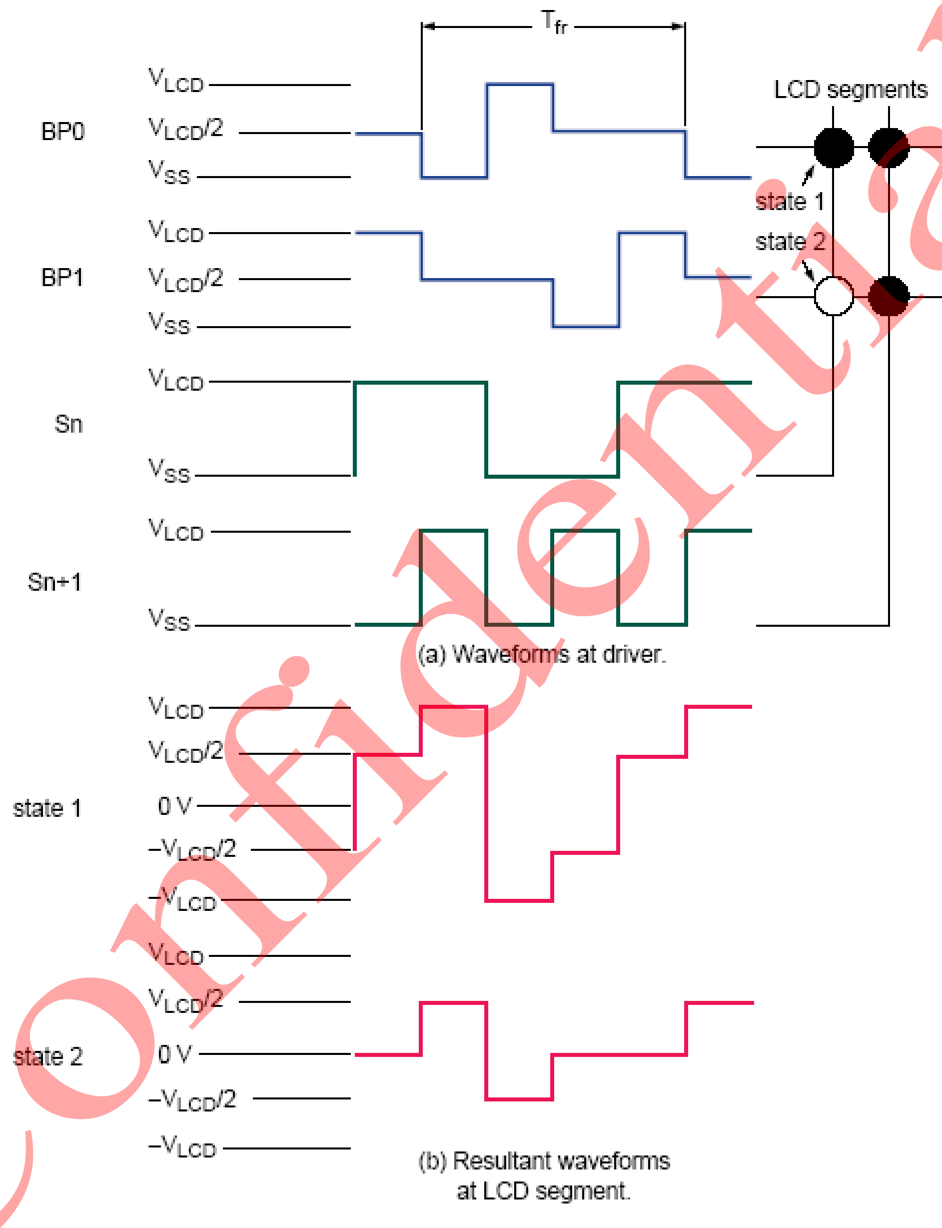
$V_{on(RMS)} = V_{LCD}$.

$V_{state2}(t) = V_{(Sn+1)}(t) - V_{BP0}(t)$.

$V_{off(RMS)} = 0$ V.

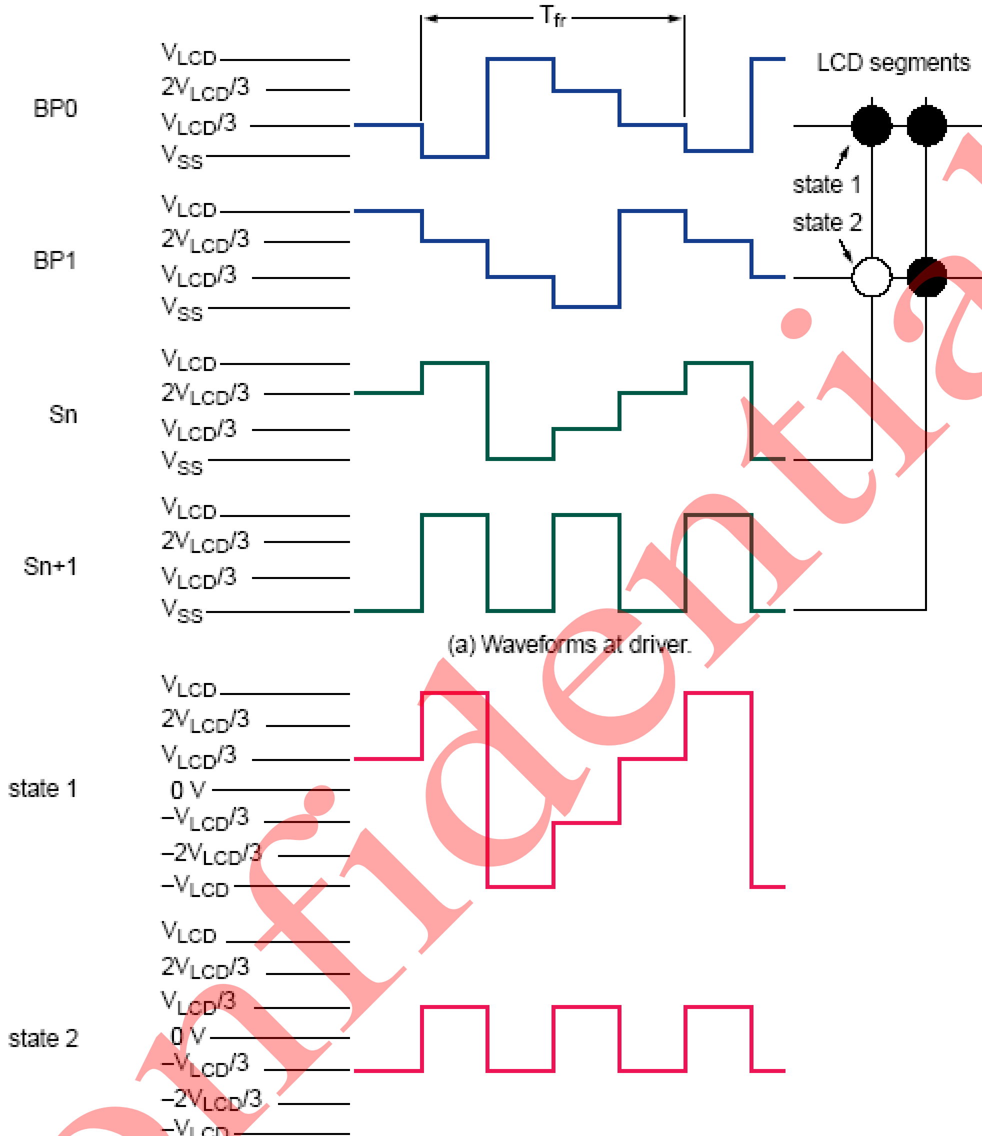
LCD Drive Mode Waveforms – 1:2 Multiplex drive mode

When two (2) backplanes are provided in the LCD, the 1:2 multiplex mode applies. The UC1672 allows the use of $\frac{1}{2}$ bias or $\frac{1}{3}$ bias in this mode as shown in the following 2 figures.



$V_{state1}(t) = V_{Sn}(t) – V_{BP0}(t)$.
$V_{on(RMS)} = 0.791V_{LCD}$.
$V_{state2}(t) = V_{Sn}(t) – V_{BP1}(t)$.
$V_{off(RMS)} = 0.354V_{LCD}$.

Waveforms for the 1:2 multiplex drive mode with $\frac{1}{2}$ bias



(a) Waveforms at driver.

(b) Resultant waveforms at LCD segment.

$V_{state1}(t) = V_{Sn}(t) – V_{BP0}(t)$.

$V_{on(RMS)} = 0.745V_{LCD}$.

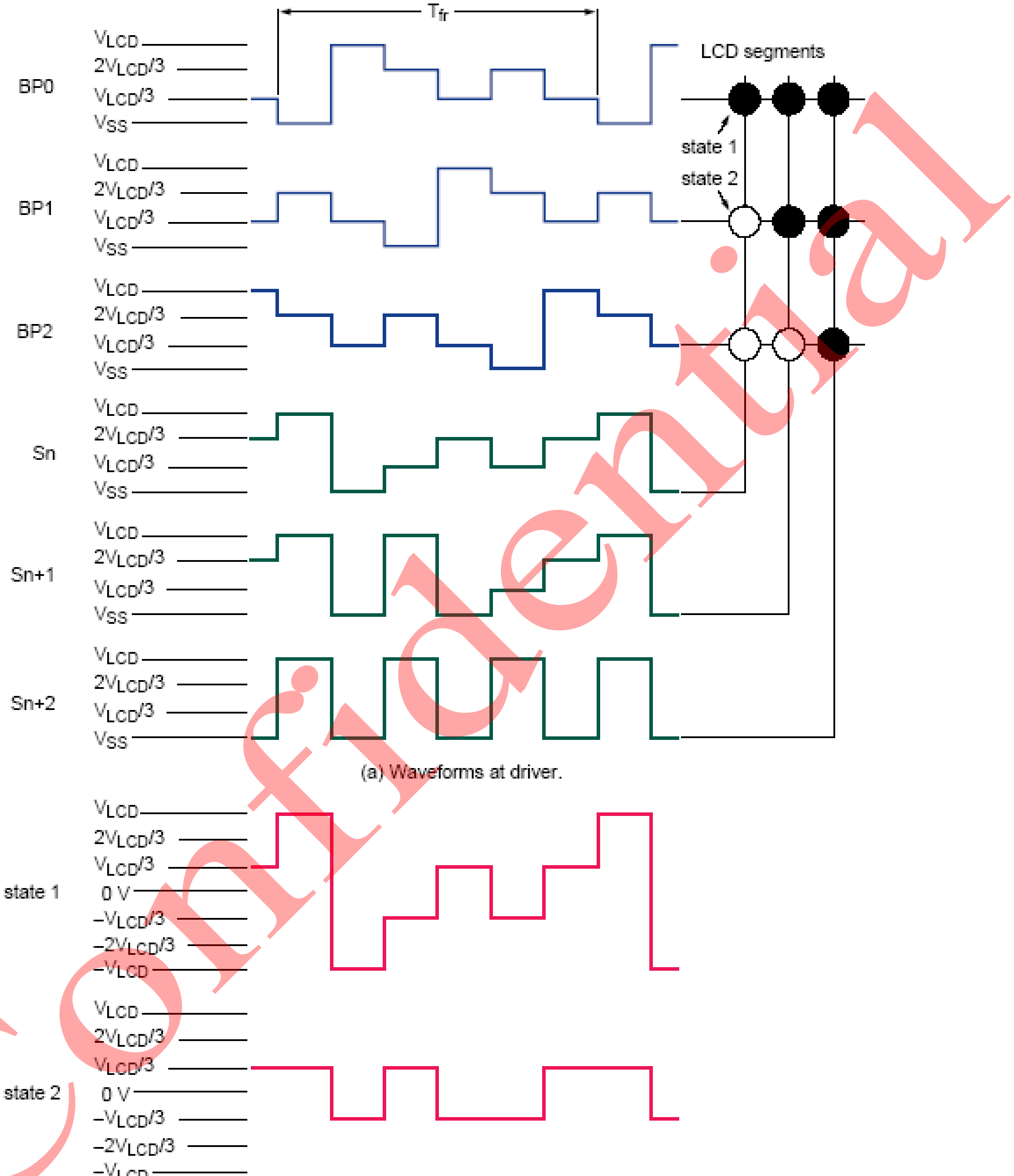
$V_{state2}(t) = V_{Sn}(t) – V_{BP1}(t)$.

$V_{off(RMS)} = 0.333V_{LCD}$.

Waveforms for the 1:2 multiplex drive mode with $\frac{1}{3}$ bias

LCD Drive Mode Waveforms – 1:3 Multiplex drive mode

When three (3) backplanes are provided in the LCD, the 1:3 multiplex drive mode applies, as shown in the figure below:



(a) Waveforms at driver.

(b) Resultant waveforms at LCD segment.

$V_{state1}(t) = V_{Sn}(t) - V_{BP0}(t)$.

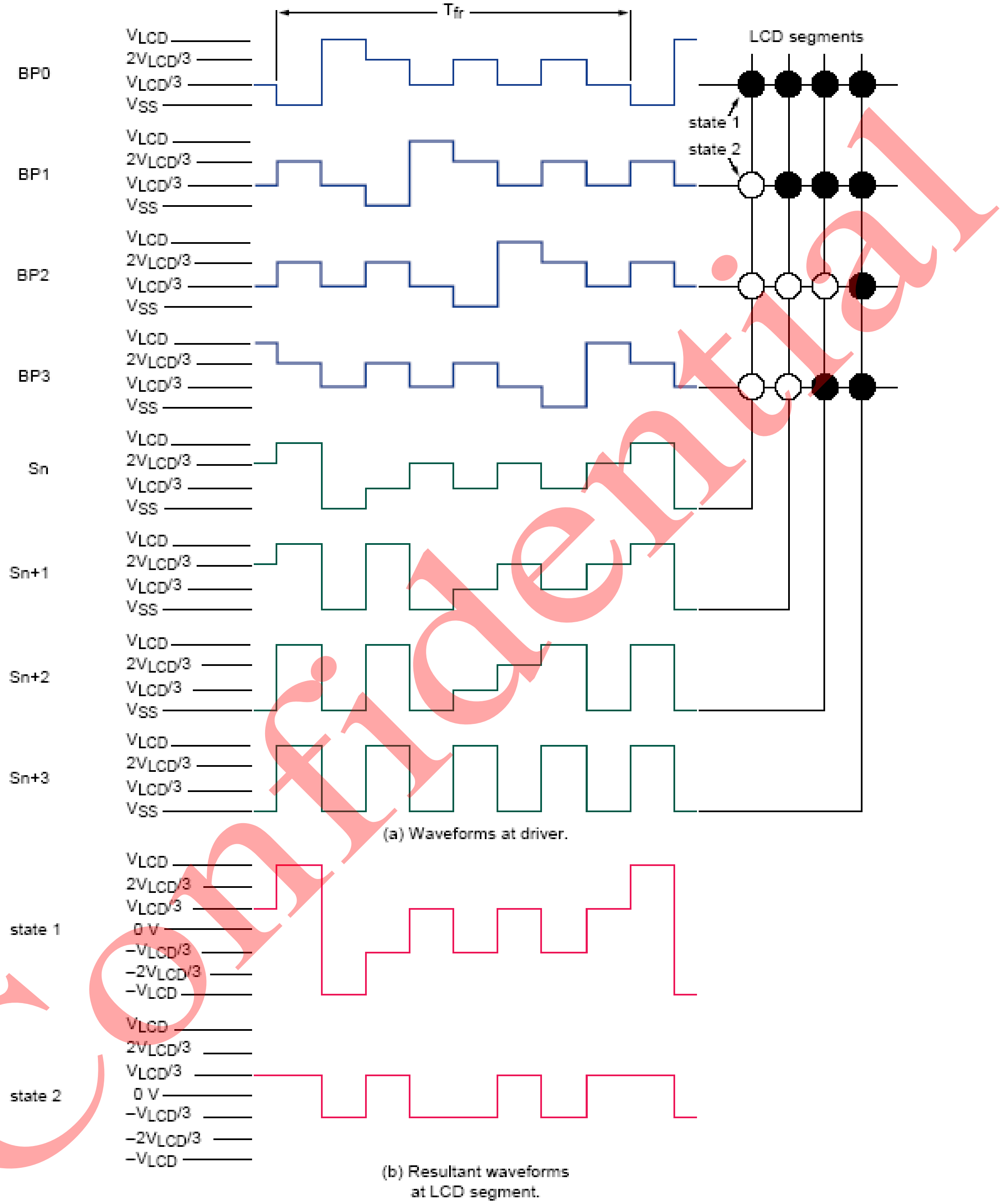
$V_{on(RMS)} = 0.638V_{LCD}$.

$V_{state2}(t) = V_{Sn}(t) - V_{BP1}(t)$.

$V_{off(RMS)} = 0.333V_{LCD}$.

LCD Drive Mode Waveforms – 1:4 Multiplex drive mode

When four (4) backplanes are provided in the LCD, the 1:4 multiplex drive mode applies, as shown in the figure below:



(a) Waveforms at driver.

(b) Resultant waveforms at LCD segment.

$V_{state1}(t) = V_{Sn}(t) - V_{BP0}(t)$.

$V_{on(RMS)} = 0.577V_{LCD}$.

$V_{state2}(t) = V_{Sn}(t) - V_{BP1}(t)$.

$V_{off(RMS)} = 0.333V_{LCD}$.

Oscillator

The internal logic and the LCD drive signals of the UC1672 are timed by a frequency f_{CLK} which either is derived from the built-in oscillator frequency f_{OSC}:

$$f_{CLK} = \frac{f_{OSC}}{64} \quad \ldots\ldots\ldots\ldots\ldots (4)$$

or equals an external clock frequency $f_{CLK(EXT)}$:

$$f_{CLK} = f_{CLK(EXT)} \quad \ldots\ldots\ldots\ldots (5)$$

Oscillator – Internal Clock

The internal oscillator is enabled by connecting pin OSC to VSS. In this case the output from pin CLK provides the clock signal for any cascaded UC1672 in the system. After power-on, pin SDI must be HIGH to guarantee that the clock starts.

Oscillator – External Clock

Connecting pin OSC to V_{DD} enables an external clock source. Pin CLK then becomes the external clock input.

Remark: A clock signal must always be supplied to the device; removing the clock may freeze the LCD in a DC state, which is not suitable for the liquid crystal.

Timing and Frame Frequency

The clock frequency (f_{CLK}) determines the LCD frame frequency (f_{FR}) and is calculated as follows:

$$f_{FR} = \frac{f_{CLK}}{24}$$

The internal clock frequency, f_{CLK}, can be selected using command Frame-Rate-select. As a result, 4 frame frequencies are available: 64Hz (typical), 82Hz, 110Hz, or 110Hz :

FR[2:0]	Typical Clock Freq.	LCD Frame Freq.
000b	1440 Hz	60 Hz
001b	1920 Hz	80 Hz
010b	2400 Hz	100 Hz
011b	2880 Hz	120 Hz
100b	3360 Hz	140 Hz
101b	3840 Hz	160 Hz
110b	4320 Hz	180 Hz
111b	4800 Hz	200Hz

However, 400Hz of frame frequency is supported when using external Oscillator, (Maximum: 9.6KHz.)

The timing of the UC1672 organizes the internal data flow of the device. This includes the transfer of display data from the display RAM to the display SEG outputs. In cascaded applications, the synchronization signal (SYNC) maintains the correct timing relationship between all the UC1672 in the system.

Display Register

The display register holds the display data while the corresponding multiplex signals are generated. There is a one-to-one relationship between the data in the display register, the LCD SEG outputs and one column of the display RAM.

Segment Outputs

The LCD drive section includes 120 SEG outputs (S0 to S119), which must be connected directly to the LCD. The SEG output signals are generated in accordance with the multiplexed backplane signals and with data residing in the display register. When less than 80 SEG outputs are required the unused SEG outputs must be left open-circuit.

Backplane Outputs

The LCD drive section includes four (4) backplane outputs: BP0 to BP3. The backplane output signals are generated in accordance with the selected LCD drive mode.

static	The same signal is carried by all four (4) backplane outputs; and they can be connected in parallel for very high drive requirements.
1:2 multiplex	BP0 and BP2, BP1 and BP3 respectively carry the same signals and can also be paired to increase the drive capabilities.
1:3 multiplex	BP3 carries the same signal as BP1 does; therefore, these two adjacent outputs can be tied together to give enhanced drive capabilities.
1:4 multiplex	BP0~ BP3 must be connected directly to the LCD.

If less than four (4) backplane outputs are required, the unused outputs can be left open-circuit.

Display RAM

The display RAM is a static 120x4-bit RAM, which stores LCD data. A logic 1 in the RAM bit map indicates the on-state of the corresponding LCD element; similarly, a logic 0 indicates the off-state. There is a one-to-one correspondence between the RAM addresses and the SEG outputs and between the individual bits of a RAM word and the backplane outputs. The display RAM bit map figure below shows rows 0 to 3, which correspond with the backplane outputs BP0 to BP3, and columns 0 to 119, which correspond with the SEG outputs S0 to S119. In multiplexed LCD applications, the SEG data of the first, second, third and fourth row of the display RAM are time-multiplexed with BP0, BP1, BP2, and BP3, respectively.

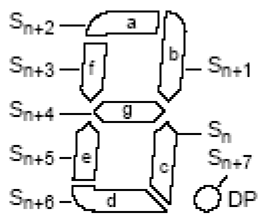
Columns→ Display RAM Addressed / SEG output (S)

Rows↓		0	1	2	3	4	…	…	…	116	117	118	119
Display RAM Rows / backplane outputs (BP)	0												
	1												
	2												
	3												

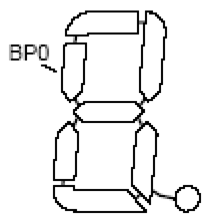
This display RAM bitmap shows the direct relationship between the display RAM addresses and the SEG outputs and between the bits in a RAM word and the backplane output.

Drive Mode: Static

LCD segments



LCD backplanes



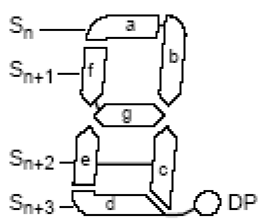
display RAM filling order — columns: display RAM address/segment outputs (s), byte1; rows: display RAM rows/backplane outputs (BP)

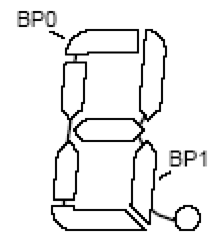
rows	n	n + 1	n + 2	n + 3	n + 4	n + 5	n + 6	n + 7
0	c	b	a	f	g	e	d	DP
1	x	x	x	x	x	x	x	x
2	x	x	x	x	x	x	x	x
3	x	x	x	x	x	x	x	x

Transmitted Display Byte (MSB … LSB):

c	b	a	f	g	e	d	DP

Drive Mode: 1:2 multiplex





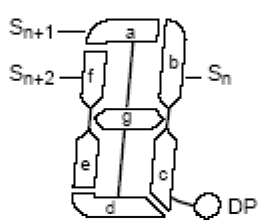
columns: display RAM address/segment outputs (s), byte1, byte2; rows: display RAM rows/backplane outputs (BP)

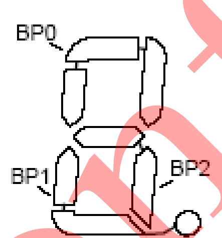
rows	n	n + 1	n + 2	n + 3
0	a	f	e	d
1	b	g	c	DP
2	x	x	x	x
3	x	x	x	x

Transmitted Display Byte:

a	b	f	g	e	c	d	DP

Drive Mode: 1:3 multiplex





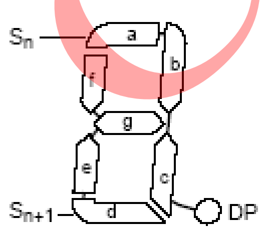
columns: display RAM address/segment outputs (s), byte1, byte2, byte3; rows: display RAM rows/backplane outputs (BP)

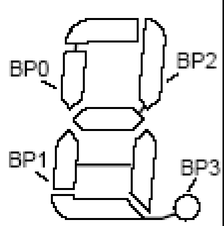
rows	n	n + 1	n + 2
0	b	a	f
1	DP	d	e
2	c	g	x
3	x	x	x

Transmitted Display Byte:

b	DP	c	a	d	g	f	e

Drive Mode: 1:4 multiplex





columns: display RAM address/segment outputs (s), byte1, byte2, byte3, byte4, byte5; rows: display RAM rows/backplane outputs (BP)

rows	n	n + 1
0	a	f
1	c	e
2	b	g
3	DP	d

Transmitted Display Byte:

a	c	b	DP	f	e	g	d

Remark: x = data bit unchanged

When display data is transmitted to the UC1672, the received display bytes are stored in the display RAM in accordance with the selected LCD drive mode. The data is stored as it arrives and does not wait for the acknowledge cycle as with the commands. Depending on the current multiplex drive mode, data is stored singularly, in pairs, triples or quadruples. To illustrate the filling order, an example of a 7-SEG display showing all drive modes is given in the Figure on the previous page; the RAM filling organization depicted applies equally to other LCD types.

The following applies to the figure on last page:

static	the 8 transmitted data bits are placed into row 0 of 8 successive 4-bit RAM words.
1:2 multiplex	the 8 transmitted data bits are placed in pairs into rows 0 and 1 of 4 successive 4-bit RAM words.
1:3 multiplex	the 8 bits are placed in triples into row 0, 1 and 2 of 3 successive 4-bit RAM words, with bit 3 of the third address left unchanged. It is not recommended to use this bit in a display because of the difficult addressing. This last bit may, if necessary, be controlled by an additional transfer to this address, but care should be taken to avoid overwriting adjacent data because always full bytes are transmitted.
1:4 multiplex	the 8 transmitted data bits are placed in quadruples into row 0, 1, 2, and 3 of 2 successive 4-bit RAM words.

Data Pointer

The addressing mechanism for the display RAM is realized using a data pointer.

This allows the loading of an individual display data byte, or a series of display data bytes, into any location of the display RAM. The sequence commences with the initialization of the data pointer by the load-data-pointer command.

Following this command, an arriving data byte is stored at the display RAM address indicated by the data pointer. The filling order is shown in the Figure on the previous page.

After each byte is stored, the content of the data pointer is automatically increased by a value dependent on the selected LCD drive mode:

Static	by eight (8)
1:2 multiplex	by four (4)
1:3 multiplex	by three (3)
1:4 multiplex	by two (2)

If an I²C-bus data access is terminated early then the state of the data pointer is unknown. Consequently, the data pointer must be rewritten prior to further RAM accesses.

Sub-address Counter

The storage of display data is conditioned by the content of the sub-address counter. Storage is allowed only when the content of the sub-address counter match with the hardware sub-address applied to A0, A1, and A2. The sub-address counter value is defined by the device-select command (see command (4) Device-select). If the content of the sub-address counter and the hardware sub-address do not match, then data storage is inhibited but the data pointer is incremented as if data storage had taken place. The sub-address counter is also increased when the data pointer overflows. The storage arrangements described lead to extremely efficient data loading in cascaded applications. When a series of display bytes are sent to the display RAM, automatic wrap-over to the next UC1672 occurs when the last RAM address is exceeded. Sub-addressing across device boundaries is successful even if the change to the next device in the cascade occurs within a transmitted character (ex. during the 27th display data byte transmitted in 1:3 mux mode).

The hardware sub-address must not be changed whilst the device is being accessed on the I²C -bus interface.

Output bank selector

The output bank selector selects one of the 4 rows per display RAM address for transfer to the display register. The actual row selected depends on the particular LCD drive mode in operation and on the instant in the multiplex sequence.

Static	Row 0 is selected
1:2 multiplex	Rows 0 and 1 are selected
1:3 multiplex	Rows 0, 1, and 2 are selected sequentially
1:4 multiplex	All RAM addresses of row 0 are selected, these are followed by the contents of row 1, row 2, and then row 3.

The UC1672 includes a RAM bank switching feature in the static and 1:2 multiplex drive modes. In the static drive mode, the bank-select command may request the contents of row 2 to be selected for display instead of the contents of row 0. In the 1:2 mode, the contents of rows 2 and 3 may be selected instead of rows 0 and 1. This gives the provision for preparing display information in an alternative bank and to be able to switch to it once it is assembled.

Input bank selector

The input bank selector loads display data into the display RAM in accordance with the selected LCD drive configuration. Display data can be loaded in row 2 in static drive mode or in rows 2 and 3 in 1:2 multiplex drive mode by using the bank-select command. The input bank selector functions independently to the output bank selector.

Blinking

The display blink capabilities of the UC1672 are very versatile. The whole display can blink at frequencies selected by the blink-select command (see command (6) Blink-select). The blink frequencies are fractions of the clock frequency. The ratios between the clock and blink frequencies depend on the blink mode selected:

Blink mode	Operating mode ratio	Blink frequency			
		fclk=1.536kHz	fclk=1.97kHz	fclk=2.640kHz	fclk=4.800kHz
off	–	blinking off	blinking off	Blinking off	Blinking off
1	fclk / 768	2.0Hz	2.5Hz	3.5Hz	6.3Hz
2	fclk / 1536	1.0Hz	1.3Hz	1.7Hz	3.1Hz
3	fclk / 3072	0.5Hz	0.6Hz	0.9Hz	1.6Hz

POWER MANAGEMENT

Power-Up Sequence for internal pump

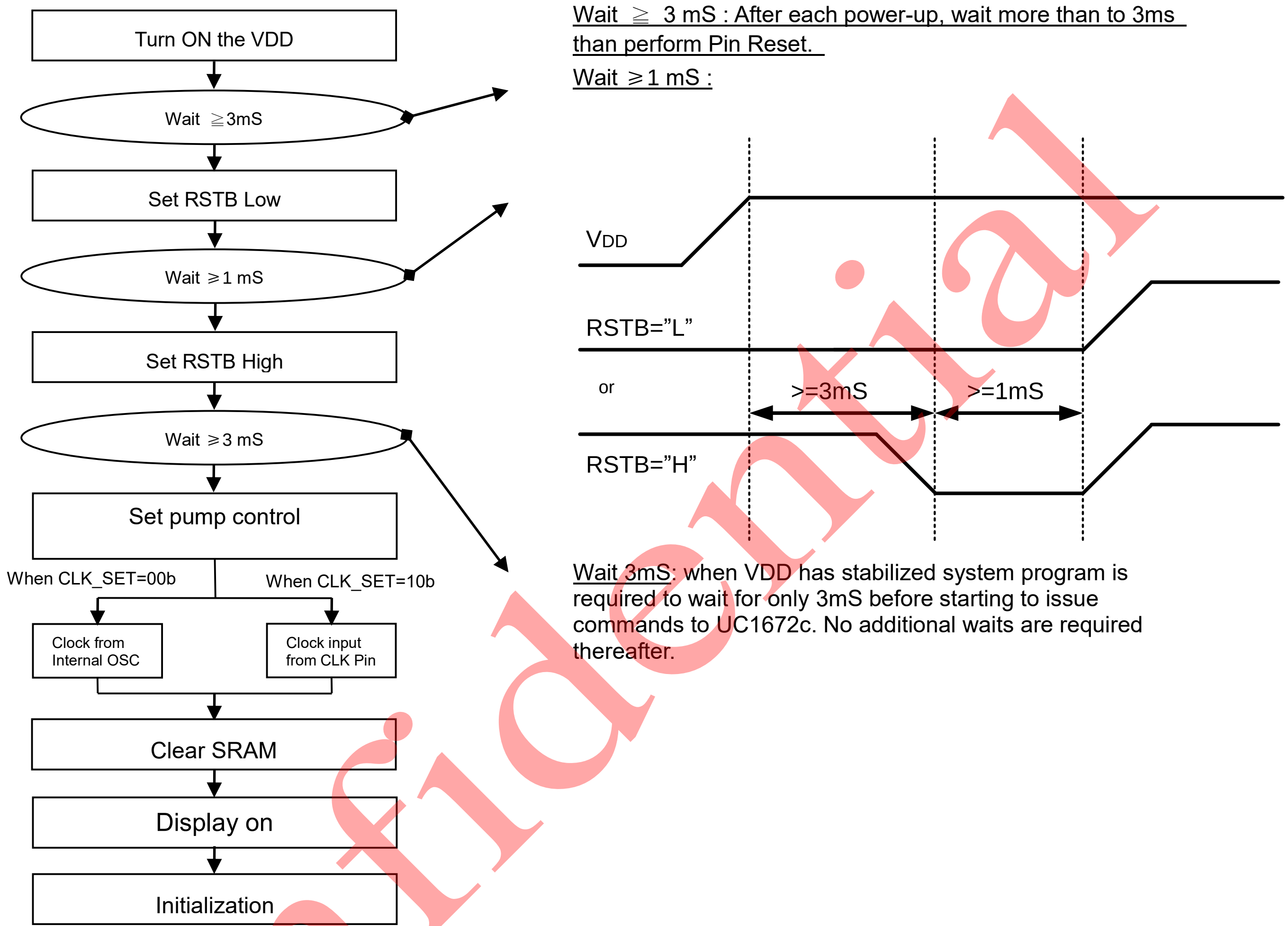


Figure 8: Reference Power-Up Sequence for internal pump

There's no delay needed while turning ON V_{DD}:

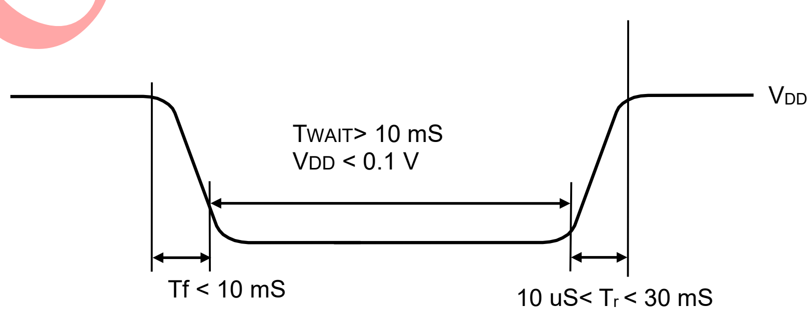


Figure 9: Power Off-On Sequence

Power-Up Sequence for external pump

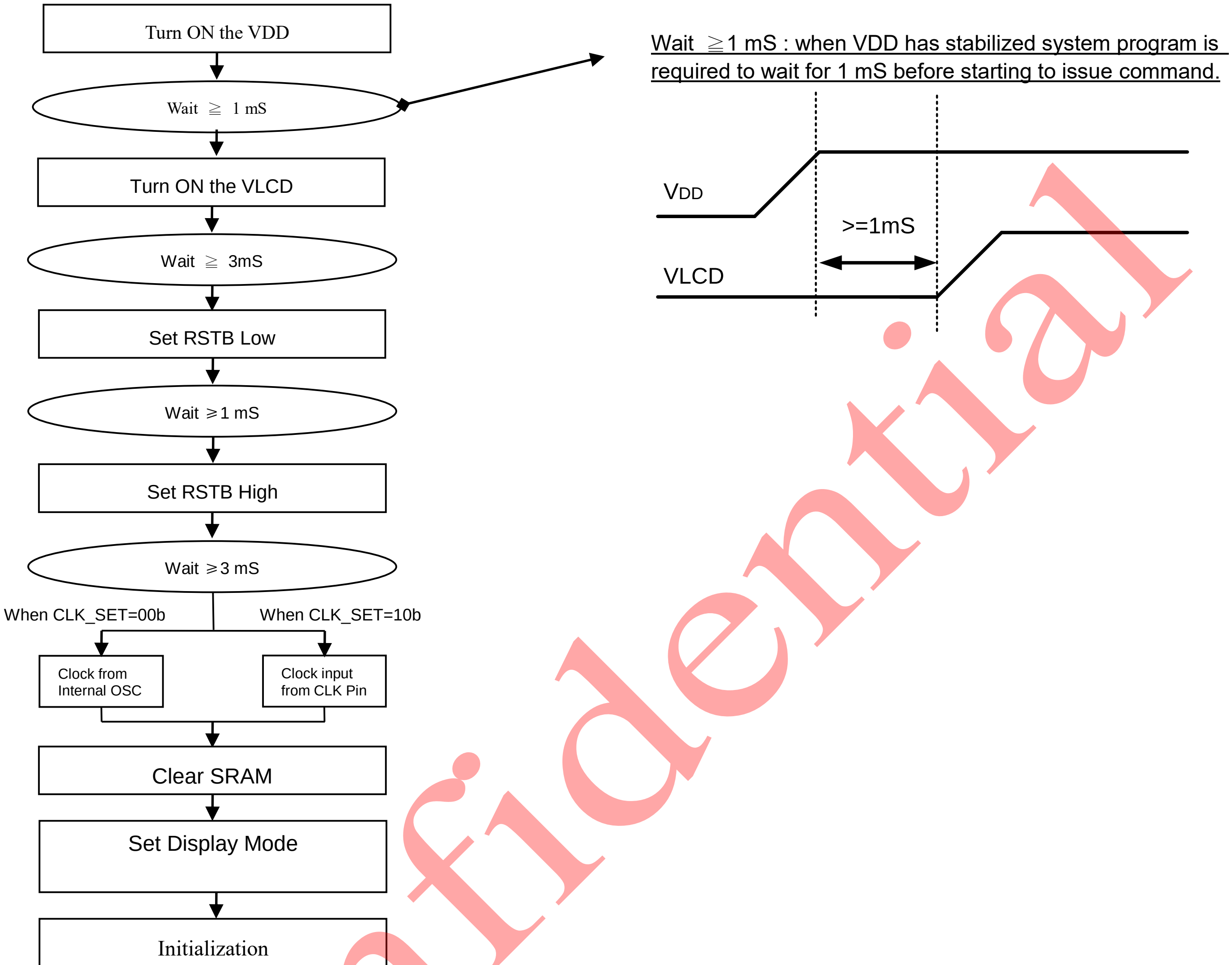
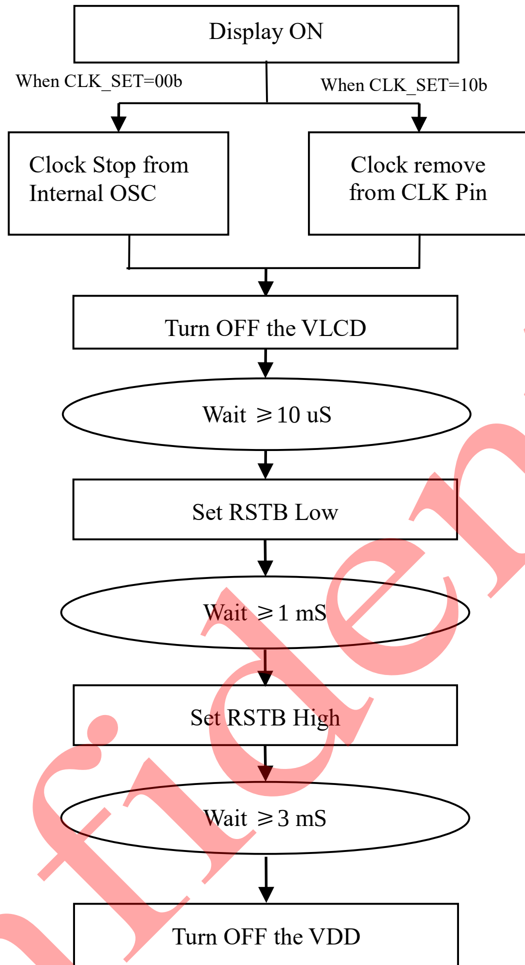


FIGURE 10: Reference Sequence – Power-Up for external pump

Power-Down Sequence

To prevent the charge stored in capacitor C_L causing abnormal residue horizontal line on display when V_{DD} is switched off, use Reset mode to enable the built-in charge draining circuit to discharge these external capacitors.



SAMPLE POWER MANAGEMENT COMMAND SEQUENCES

The following tables are examples of command sequence for power-up, power-down and display ON/OFF operations. These are only to demonstrate some “*typical, generic*” scenarios. Designers are encouraged to study related sections of the datasheet and find out what the best parameters and control sequences are for their specific design needs.

Type Required: These items are required
Customized: These items are not necessary if customer parameters are the same as default
Advanced: We recommend new users to skip these commands and use default values.
Optional: These commands depend on what users want to do.

C/D The type of the interface cycle. It can be either Command (0) or Data (1)

W/R The direction of data flow of the cycle. It can be either Write (0) or Read (1).

Power-Up

Type	C/D	D7	D6	D5	D4	D3	D2	D1	D0	Chip action	Comments
R	-	–	–	–	–	–	–	–	–	Set RSTB pin Low	Wait 1mS after RSTB is Low
R		–	–	–	–	–	–	–	–	Set RSTB pin High	Wait 3mS after RSTB is High
R	0	0	1	0	1	0	0	0	1	Select Device	
R	1	1	1	1	0	0	0	0	0		
R	0	0	1	0	1	0	0	1	0	Select Frame Rate & Blink	
R	1	1	0	0	0	0	0	1	0		
R	0	0	1	0	1	0	0	1	1	Set pump control	
R	1	0	0	0	0	1	1	0	1		
R	0	0	1	0	1	0	1	0	0	Set pump voltage	
R	1	0	0	1	1	0	0	1	0		
R	0	0	1	0	1	0	1	1	0	Set RAM address	
R	1	0	0	0	0	0	0	0	0		
O	1	# . . #	# . . #	# . . #	# . . #	# . . #	# . . #	# . . #	# . . #	Write display RAM Bit Map	Set up display image
R	0	0	1	0	1	0	1	0	1	Set Display Mode	Set display enable
R	1	0	0	0	0	0	0	0	1		

Power-Down

Type	C/D	D7	D6	D5	D4	D3	D2	D1	D0	Chip action	Comments
R	0	0	1	0	1	0	1	0	1	Set Display Mode	Set display disable
R	1	0	0	0	0	0	0	0	0		
R		–	–	–	–	–	–	–	–		VDD and VLCD OFF

ABSOLUTE MAXIMUM RATING

In accordance with the Absolute Maximum Rating System (IEC 134).

Symbol	Parameter	Min.	Max.	Unit
V_{DD}	Logic Supply voltage	-0.3	+5.5	V
V_{LCD}	LCD Generated voltage	-0.3	+5.5	V
V_{IN} / V_{OUT}	Any input/output	-0.4	V_{DD} + 0.3	V
T_{OPR}	Operating temperature range	-30	+85	ºC
T_{STR}	Storage temperature	-55	+125	ºC

Note:

1. V_{DD} is based on V_{SS} = 0V.
2. Stress beyond ranges listed above may cause permanent damage to the device. These are stress rating only. This device should be operated under DC/AC characteristics condition for normal operation. Exposure to over the DC/AC characteristics conditions for extended periods may affect device reliability and function operation.

DC CHARACTERISTICS

V_{DD}=3V~5.5V, V_{SS}=0V, V_{LCD}=2.5V~5.5V, Ta=–30°C~+85°C, unless otherwise specified.

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
Supply						
V_{DD}	Supply voltage		3	-	5.5	V
V_{LCD}	LCD supply voltage		2.5		5.5	V
IDD(LCD)	LCD voltage current	$f_{CLK(EXT)}$=1536Hz [1]		16	60	uA
IDD	Supply current	$f_{CLK(EXT)}$=1536Hz [2]		2	20	uA
Logic						
V_I	Input voltage		Vss-0.5	-	V_{DD}+0.5	V
V_{IH}	Input voltage HIGH	on pins CLK, $\overline{\text{SYNC}}$,OSC,A0~A2,SA0	0.8V_{DD}		V_{DD}	V
V_{IL}	Input voltage LOW		Vss		0.2V_{DD}	V
V_{OH}	Output voltage HIGH		0.8V_{DD}	-	-	V
V_{OL}	Output voltage LOW		-	-	0.2V_{DD}	V
I_{OH}	Output current HIGH	on pins CLK, V_{OH}=4.6V,V_{DD}=5V	+1	-	-	mA
I_{OL}	Output current LOW	on pins CLK, $\overline{\text{SYNC}}$, V_{OL}=0.4V,V_{DD}=5V	-	-	-1	mA
I_L	Leakage current	on pins OSC,CLK,SCL,SDI,A0~A2,SA0 V_I=V_{DD} or Vss	-1	-	+1	µA
C_I	Input capacitance	[2]	-	-	7	pF
I²C bus, input on pins SDI and SCL						
V_I	Input voltage		Vss-0.5	-	5.5	V
V_{IH}	Input voltage HIGH		0.8V_{DD}	-	5.5	V
V_{IL}	Input voltage LOW		Vss	-	0.2V_{DD}	V
C_I	Input capacitance	[2]	-	-	7	pF
I_{OL}(SDI)	Output current LOW	V_{OL}=0.4V, V_{DD}=5V, on pin SDI	+3	-	-	mA
LCD outputs						
ΔV_O	Output voltage variaition	C_{BPL}=35nF, on pin BPx	-100	-	+100	mV
		C_{SGM}=5nF, on pin Sx	-100	-	+100	mV
R_O	Output resistance	V_{LCD} = 5V, on pin BPx [3]	-	1.5	10	kΩ
		V_{LCD}= 5V, on pin Sx [3]		6.0	13.5	kΩ

Note:
[1] LCD outputs are open-circuit; inputs at Vss or V_{DD}; external clock with 50% duty factor; I²C bus inactive.
[2] Not tested, design specification only.
[3] Outputs measured individually and sequentially.

Power Consumption

V_{DD} =3.3 V V_{LCD} = 5 V, ,FR = 100Hz,

Display Pattern	Conditions	Typ.	Max.	Unit
All-OFF	Bus = idle	256	384	µA
All-ON	Bus = idle	261	391	µA
1-pixel checker	Bus = idle	263	394	µA
--	Bus = idle (standby current)	-	5.5	µA

AC CHARACTERISTICS

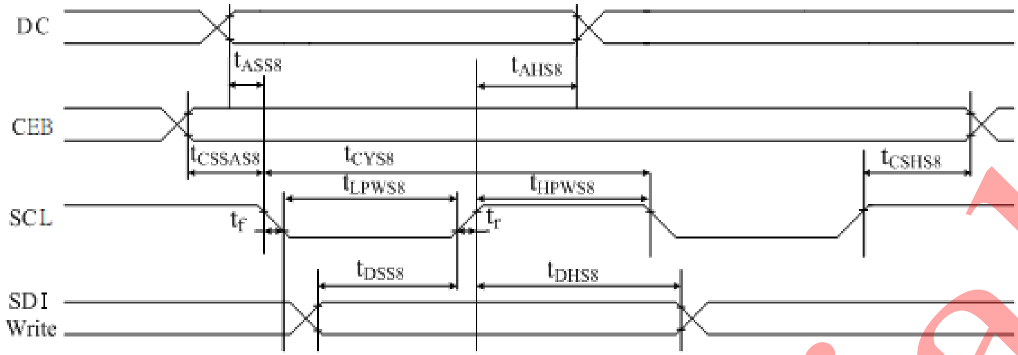


FIGURE 14: Serial Bus Timing Characteristics (for S8)

Symbol	Signal	Description	Condition	Min.	Max.	Units
(3V ≤ V_{DD} < 5.5V, Ta= –30 to +85°C)				(Read / Write)		
t_{ASS8} t_{AHS8}	DC	Address setup time Address hold time		0 30	–	nS
t_{CSSAS8} t_{CSHS8}	CEB	Chip select setup time Chip select hold time		5 5	–	nS
t_{CYS8}	SCL	Cycle time	VDD=3V VDD=5V	170 100	–	nS
t_{LPWS8}		Low pulse width	VDD=3V VDD=5V	80 50		
t_{HPWS8}		High pulse width	VDD=3V VDD=5V	90 40		
t_{DSS8}	SDI (Write)	Data setup time	VDD=3V VDD=5V	40 20	–	nS
t_{DHS8}		Data hold time	VDD=3V VDD=5V	50 30		

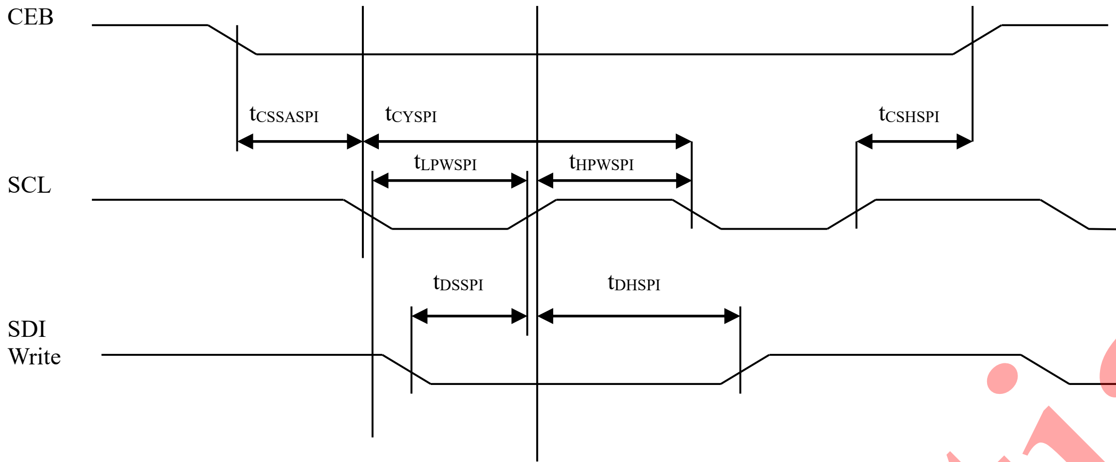


FIGURE 15: Serial Bus Timing Characteristics (for Standard SPI)

Symbol	Signal	Description	Condition	Min.	Max.	Unit
($3V \leq V_{DD} < 5.5V$, Ta= –30 to +85°C)				(Read / Write)		
$t_{CSSASPI}$ t_{CSHSPI}	CEB	Chip select setup time		5 5	–	nS
t_{CYSPI}	SCL	Cycle time	VDD=3V VDD=5V	170 100	–	nS
t_{LPWSPI}		Low pulse width	VDD=3V VDD=5V	80 50		
t_{HPWSPi}		High pulse width	VDD=3V VDD=5V	90 40		
t_{DSSPI}	SDI (Write)	Data setup time	VDD=3V VDD=5V	40 20	–	nS
t_{DHSPI}		Data hold time	VDD=3V VDD=5V	50 30		

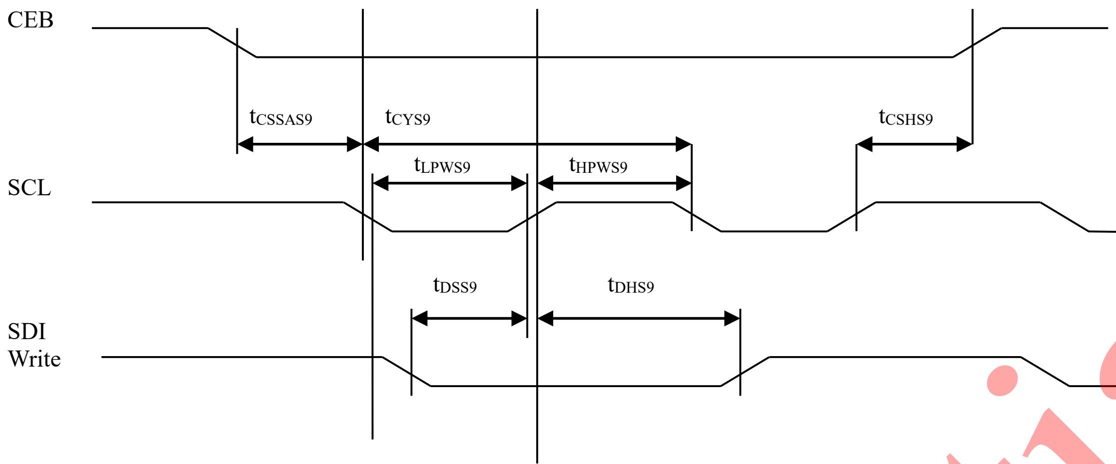


FIGURE 16: Serial Bus Timing Characteristics (for S9)

Symbol	Signal	Description	Condition	Min.	Max.	Unit
($3V \le V_{DD} < 5.5V$, Ta= –30 to +85°C)				(Read / Write)		
t_{CSSAS9} t_{CSHS9}	CEB	Chip select setup time		5 5	–	nS
t_{CYS9}	SCL	Cycle time	VDD=3V VDD=5V	170 100	–	nS
t_{LPWS9}		Low pulse width	VDD=3V VDD=5V	80 50		
t_{HPWS9}		High pulse width	VDD=3V VDD=5V	90 40		
t_{DSS9}	SDI (Write)	Data setup time	VDD=3V VDD=5V	40 20	–	nS
t_{DHS9}		Data hold time	VDD=3V VDD=5V	50 30		

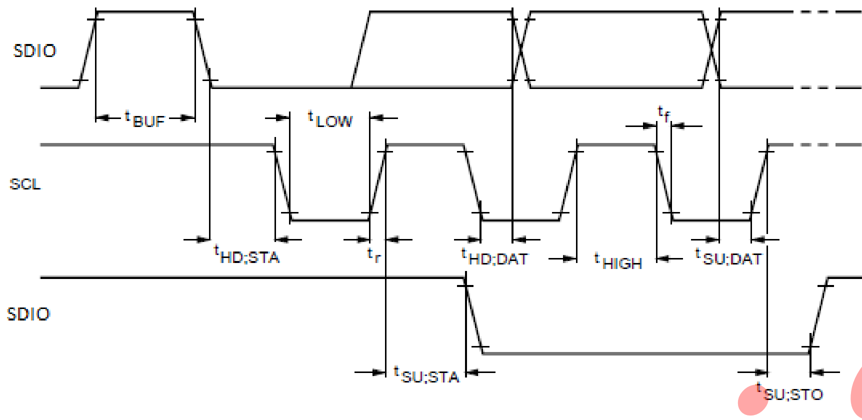


Figure: I²C Series Interface Characteristics

3V ≤ V_{DD} < 5.5V, Ta= –30 to +85°C

Symbol	Signal	Description	Condition	Min.	Typ.	Max.	Unit
I2C bus timing							
f_{SCL}	SCL	SCL clock frequency		–	–	400	kHz
t_{HIGH}	SCL	HIGH period of the SCL clock		0.6	–	–	uS
t_{LOW}	SCL	LOW period of the SCL clock		1.3	–	–	uS
SDIO pin							
$t_{SU/DAT}$	SDIO	Data set-up time		180	–	–	nS
$t_{HD;DAT}$	SDIO	Data hold time		0	–	–	nS
SCL and SDIO pins							
Tbuf	SCL, SDIO	Bus free time between STOP & START		1.3	–	–	uS
Tsu;sto	SCL, SDIO	Set-up time from STOP		0.6	–	–	uS
Thd;sta	SCL, SDIO	Hold time (repeated) START		0.6	–	–	uS
Tsu;sta	SCL, SDIO	Set-up time for a repeated START		0.6	–	–	uS
Tr	SCL, SDIO	Rise time of both SDIO and SCL signals		–	–	0.3	uS
Tf	SCL, SDIO	Fall time of both SDIO and SCL signals		–	–	0.3	uS
Cb	SCL, SDIO	Capacitive load for each bus line		–	–	400	pF
Tw(spike)	SCL, SDIO	Spike pulse width	On bus	–	–	50	nS

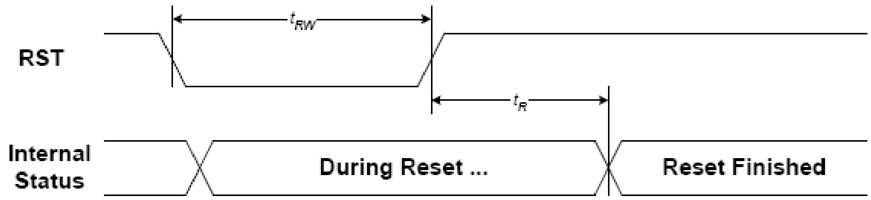
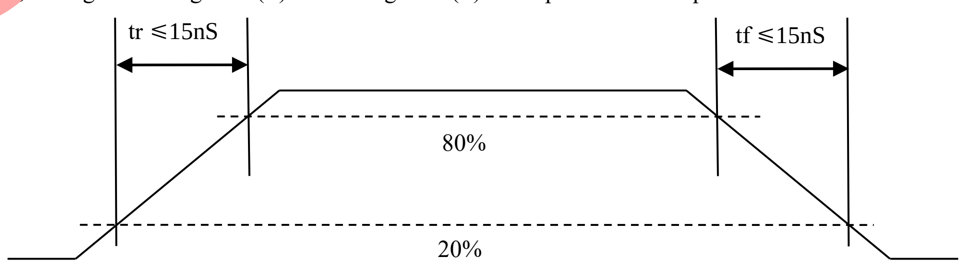


FIGURE 15: Reset Characteristics

(3V ≤ V_{DD} < 5.5V, Ta= –30 to +85°C)

Symbol	Signal	Description	Condition	Min.	Max.	Units
t_{RW}	RSTB	Reset low pulse width		1	–	mS
t_R	RSTB Internal Status	Reset to Internal Status pulse delay		3	–	mS

Note: For each mode, the signal's rising time (tr) and falling time (tf) are stipulated to be equal to or less than 15nS each.



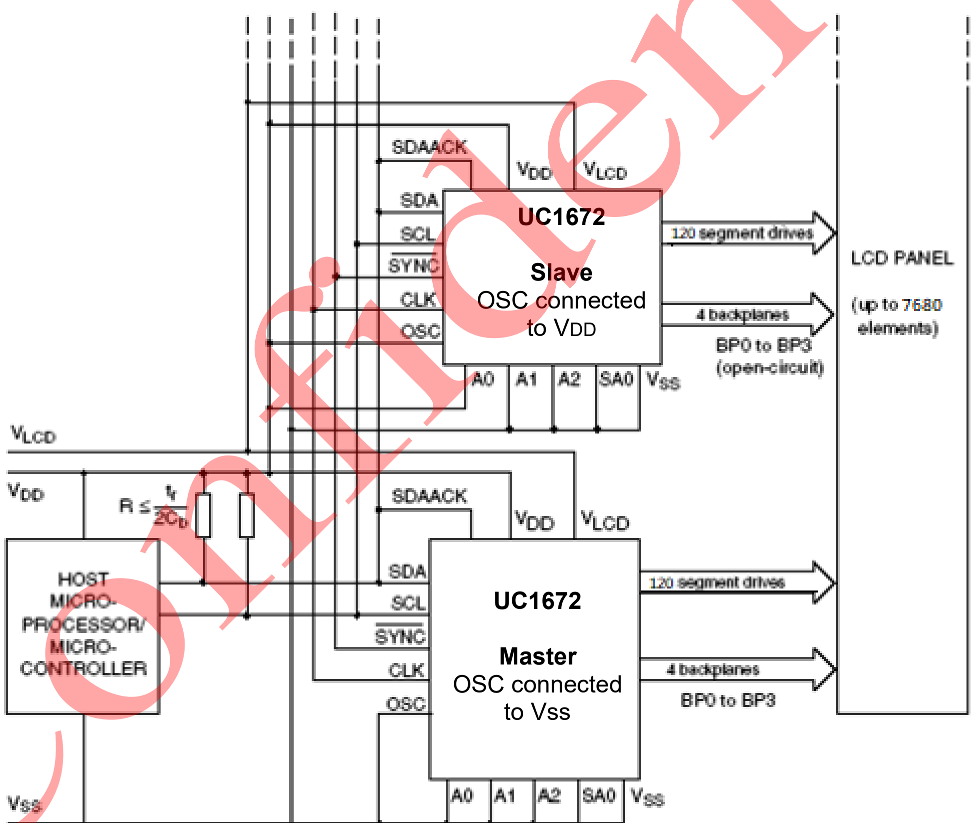
CASCADED OPERATION

In large display configurations, up to 16 UC1672 can be recognized on the same I²C bus by using the 3-bit hardware subaddress (A0, A1, and A2) and the programmable I²C bus slave address (SA0).

Cluster	SA0	A[2:0]	Device
1	0	000	0
		001	1
		010	2
		011	3
		100	4
		101	5
		110	6
		111	7

Cluster	SA0	A[2:0]	Device
2	1	000	8
		001	9
		010	10
		011	11
		100	12
		101	13
		110	14
		111	15

When cascaded UC1672 are synchronized, they can share the backplane signals from one of the devices in the cascade. Such an arrangement is cost-effective in large LCD applications since the backplane outputs of only one device need to be through-plated to the backplane electrodes of the display. The other UC1672 of the cascade contribute additional SEG outputs, but their backplane outputs are left open-circuit (See the following Figure).



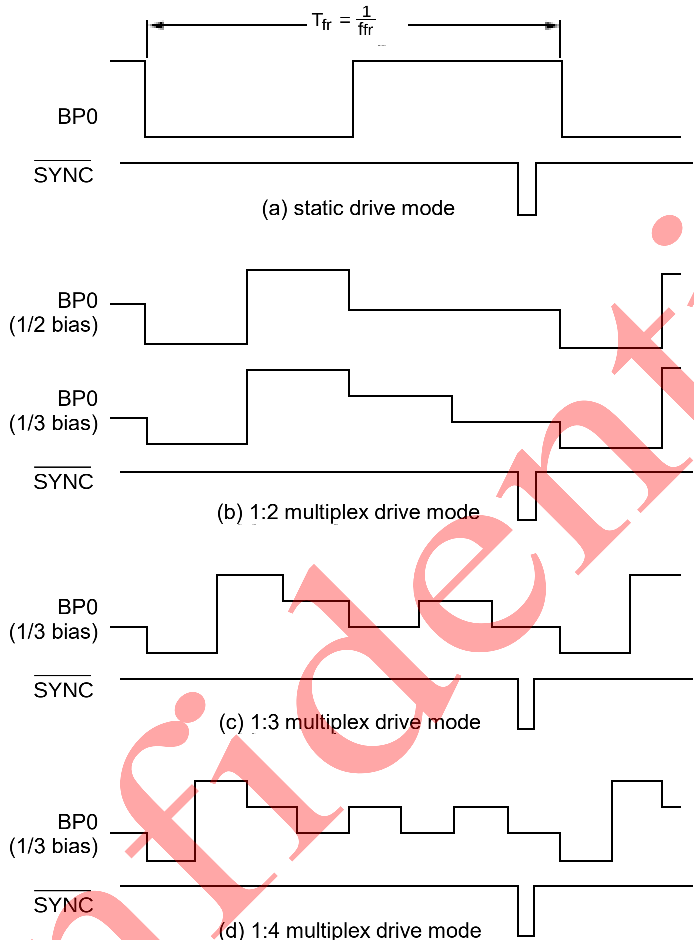
For display sizes that are not multiple of 480 elements, a mixed cascaded system can be considered containing only devices like UC1672. Depending on the application, one must take care of the software command and pin connection compatibility.

Only one master but multiple slaves are allowed in a cascade. No external clock should be used; the slaves get the clock from the master.

The $\overline{SYNC}$ line is provided to maintain the correct synchronization between all cascaded UC1672's. The only time that $\overline{SYNC}$ is likely to be needed is if synchronization is accidentally lost (e.g. by noise in adverse electrical environments or by the definition of a multiplex drive mode when UC1672 with different SA0 levels are cascaded).

$\overline{SYNC}$ is organized as an input/output pin; The output selection is realized as an open-drain driver with an internal pull-up resistor. A UC1672 asserts the $\overline{\text{SYNC}}$ line at the onset of its last active backplane signal and monitors the $\overline{\text{SYNC}}$ line at all other times.

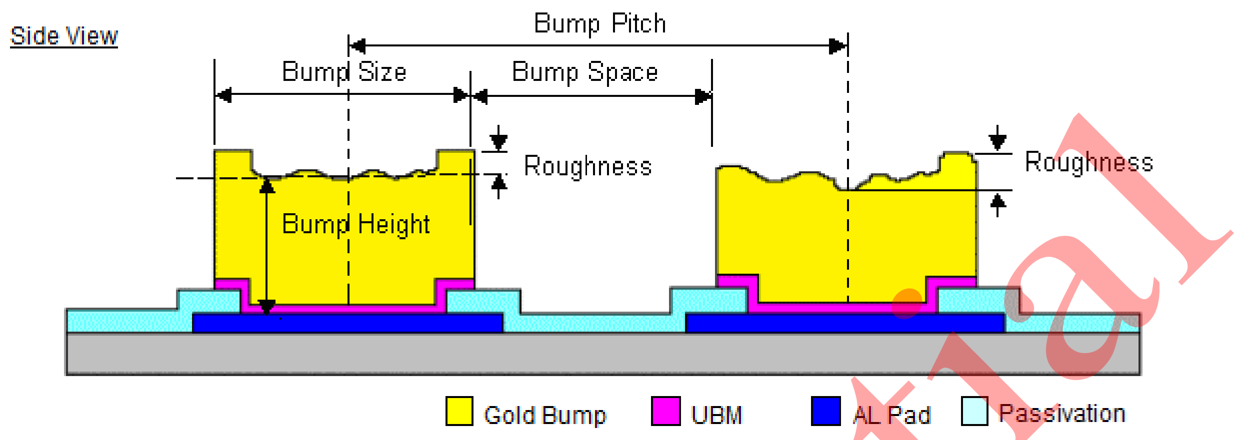
If synchronization in the cascade is lost, it is restored by the first UC1672 to assert $\overline{\text{SYNC}}$. The timing relationships between the backplane waveforms and the $\overline{\text{SYNC}}$ signal for the various drive modes of the UC1672 are shown in the Figure below.

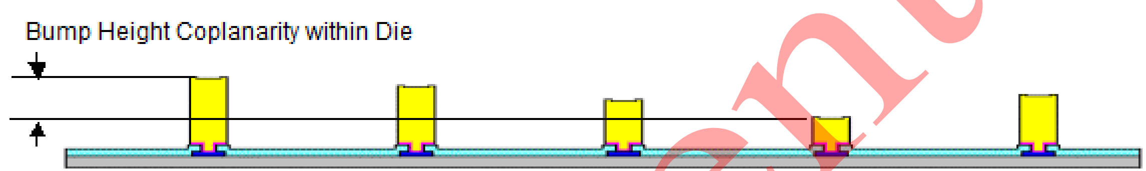


The contact resistance between the SYNC pins of cascaded devices must be controlled. If the resistance is too high, then the device will not be able to synchronize properly. This is particularly applicable to COG applications. The table below shows the limited values for contact resistance.

Number of devices	Maximum contact resistance
2	6000 Ω
3~5	2200 Ω
6~10	1200 Ω
11~16	700 Ω

PHYSICAL DIMENSIONS

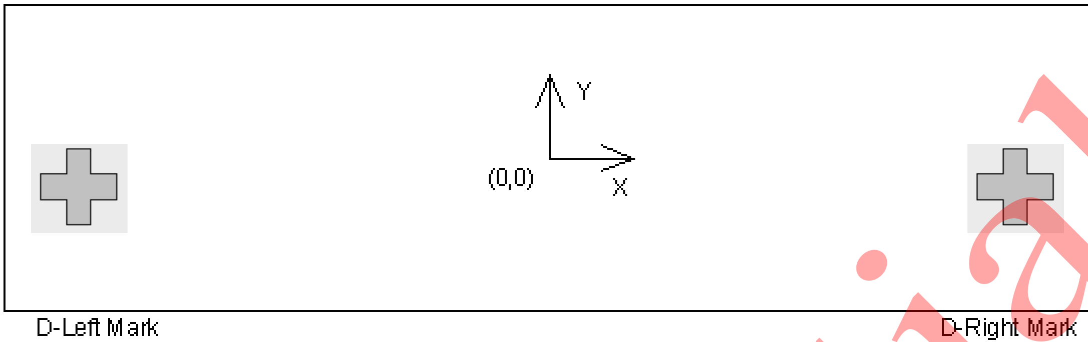




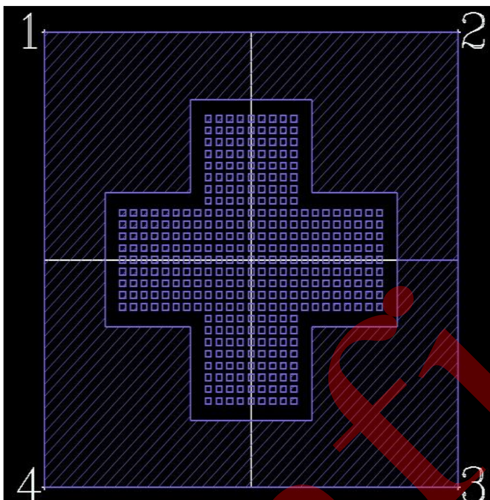
Chip Size:	**(4038± 40)um x (938 ± 40)um**
Die Thickness:	**400um ± 20um**
	Die TTV≦2um
Bump Height:	**12 ± 3um,**
	Hmax - Hmin ≦2um
Bump Size:	**15um x 100 um ± 2um**
Bump Area:	**1500 um^2**
Bump Pitch:	**30um**
Bump Gap:	**15um ± 3um**
Hardness:	**90Hv ± 25Hv,**
	Shear≧5g/mil2

ALIGNMENT MARK INFORMATION

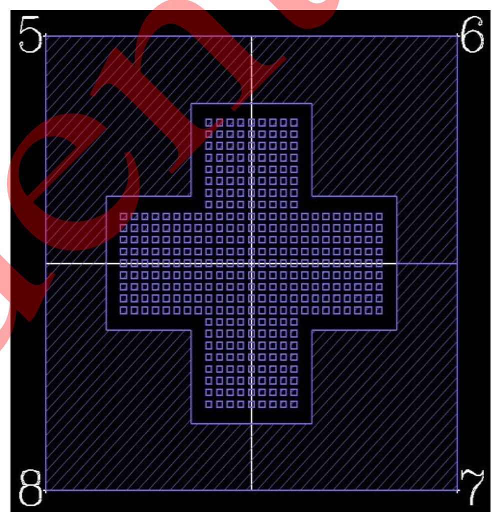
ALIGNMENT MARK POSITION:



ALIGNMENT MARK SHAPES:



Left mark



Right mark

ALIGNMENT MARK COORDINATES:

Point	Left mark		Point	Right mark	
	X	Y		X	Y
Center_ Left	-1897.5	-366.5	Center_ Right	1897.5	-366.5
1	-1940	-324	5	1855	-324
2	-1855	-324	6	1940	-324
3	-1855	-409	7	1940	-409
4	-1940	-409	8	1855	-409

PAD COORDINATES

No	Name	X	Y	W	H
1	NC<0>	-1965	-382	15	100
2	VLCD	-1790	-393	40	40
3	VLCD	-1730	-393	40	40
4	VLCD	-1640	-393	40	40
5	VLCD	-1580	-393	40	40
6	VLCD	-1520	-393	40	40
7	VLCD	-1460	-393	40	40
8	VSSA	-1400	-393	40	40
9	VSSA	-1340	-393	40	40
10	VSSA	-1280	-393	40	40
11	VSSA	-1220	-393	40	40
12	VSS	-1160	-393	40	40
13	VSS	-1100	-393	40	40
14	VSS	-1040	-393	40	40
15	VSS	-980	-393	40	40
16	VSS	-920	-393	40	40
17	VSS	-860	-393	40	40
18	VSS	-800	-393	40	40
19	VSS	-740	-393	40	40
20	VSS	-680	-393	40	40
21	VDD	-620	-393	40	40
22	VDD	-560	-393	40	40
23	TO<0>	-500	-393	40	40
24	TO<1>	-440	-393	40	40
25	TO<2>	-380	-393	40	40
26	TO<3>	-320	-393	40	40
27	TO<4>	-260	-393	40	40
28	TO<5>	-200	-393	40	40
29	TI<0>	-140	-393	40	40
30	TI<1>	-80	-393	40	40
31	SA0	-20	-393	40	40
32	IFS0	40	-393	40	40
33	IFS1	100	-393	40	40
34	A0	160	-393	40	40
35	A1	220	-393	40	40
36	A2	280	-393	40	40

No	Name	X	Y	W	H
37	VDD	340	-393	40	40
38	VDD	400	-393	40	40
39	RSTEN	460	-393	40	40
40	RSTB	520	-393	40	40
41	TAO	580	-393	40	40
42	CLK	640	-393	40	40
43	VSS	700	-393	40	40
44	VSS	760	-393	40	40
45	EVENT	820	-393	40	40
46	SDO	880	-393	40	40
47	SDI	944	-393	40	40
48	DC	1004	-393	40	40
49	SCL	1064	-393	40	40
50	CEB	1124	-393	40	40
51	SYNC	1184	-393	40	40
52	FSOURCE	1244	-393	40	40
53	FSOURCE	1304	-393	40	40
54	NC	1408	-393	40	40
55	NC	1468	-393	40	40
56	VSS	1572	-393	40	40
57	VSS	1632	-393	40	40
58	VLCD	1730	-393	40	40
59	VLCD	1790	-393	40	40
60	NC<1>	1965	-382	15	100
61	NC<2>	1965	389	15	100
62	BP_R<0>	1935	389	15	100
63	BP_R<1>	1905	389	15	100
64	BP_R<2>	1875	389	15	100
65	BP_R<3>	1845	389	15	100
66	SEG<119>	1785	389	15	100
67	SEG<118>	1755	389	15	100
68	SEG<117>	1725	389	15	100
69	SEG<116>	1695	389	15	100
70	SEG<115>	1665	389	15	100
71	SEG<114>	1635	389	15	100
72	SEG<113>	1605	389	15	100

No	Name	X	Y	W	H
73	SEG<112>	1575	389	15	100
74	SEG<111>	1545	389	15	100
75	SEG<110>	1515	389	15	100
76	SEG<109>	1485	389	15	100
77	SEG<108>	1455	389	15	100
78	SEG<107>	1425	389	15	100
79	SEG<106>	1395	389	15	100
80	SEG<105>	1365	389	15	100
81	SEG<104>	1335	389	15	100
82	SEG<103>	1305	389	15	100
83	SEG<102>	1275	389	15	100
84	SEG<101>	1245	389	15	100
85	SEG<100>	1215	389	15	100
86	SEG<99>	1185	389	15	100
87	SEG<98>	1155	389	15	100
88	SEG<97>	1125	389	15	100
89	SEG<96>	1095	389	15	100
90	SEG<95>	1065	389	15	100
91	SEG<94>	1035	389	15	100
92	SEG<93>	1005	389	15	100
93	SEG<92>	975	389	15	100
94	SEG<91>	945	389	15	100
95	SEG<90>	915	389	15	100
96	SEG<89>	885	389	15	100
97	SEG<88>	855	389	15	100
98	SEG<87>	825	389	15	100
99	SEG<86>	795	389	15	100
100	SEG<85>	765	389	15	100
101	SEG<84>	735	389	15	100
102	SEG<83>	705	389	15	100
103	SEG<82>	675	389	15	100
104	SEG<81>	645	389	15	100
105	SEG<80>	615	389	15	100
106	SEG<79>	585	389	15	100
107	SEG<78>	555	389	15	100
108	SEG<77>	525	389	15	100
109	SEG<76>	495	389	15	100
110	SEG<75>	465	389	15	100

No	Name	X	Y	W	H
111	SEG<74>	435	389	15	100
112	SEG<73>	405	389	15	100
113	SEG<72>	375	389	15	100
114	SEG<71>	345	389	15	100
115	SEG<70>	315	389	15	100
116	SEG<69>	285	389	15	100
117	SEG<68>	255	389	15	100
118	SEG<67>	225	389	15	100
119	SEG<66>	195	389	15	100
120	SEG<65>	165	389	15	100
121	SEG<64>	135	389	15	100
122	SEG<63>	105	389	15	100
123	SEG<62>	75	389	15	100
124	SEG<61>	45	389	15	100
125	SEG<60>	15	389	15	100
126	SEG<59>	-15	389	15	100
127	SEG<58>	-45	389	15	100
128	SEG<57>	-75	389	15	100
129	SEG<56>	-105	389	15	100
130	SEG<55>	-135	389	15	100
131	SEG<54>	-165	389	15	100
132	SEG<53>	-195	389	15	100
133	SEG<52>	-225	389	15	100
134	SEG<51>	-255	389	15	100
135	SEG<50>	-285	389	15	100
136	SEG<49>	-315	389	15	100
137	SEG<48>	-345	389	15	100
138	SEG<47>	-375	389	15	100
139	SEG<46>	-405	389	15	100
140	SEG<45>	-435	389	15	100
141	SEG<44>	-465	389	15	100
142	SEG<43>	-495	389	15	100
143	SEG<42>	-525	389	15	100
144	SEG<41>	-555	389	15	100
145	SEG<40>	-585	389	15	100
146	SEG<39>	-615	389	15	100
147	SEG<38>	-645	389	15	100
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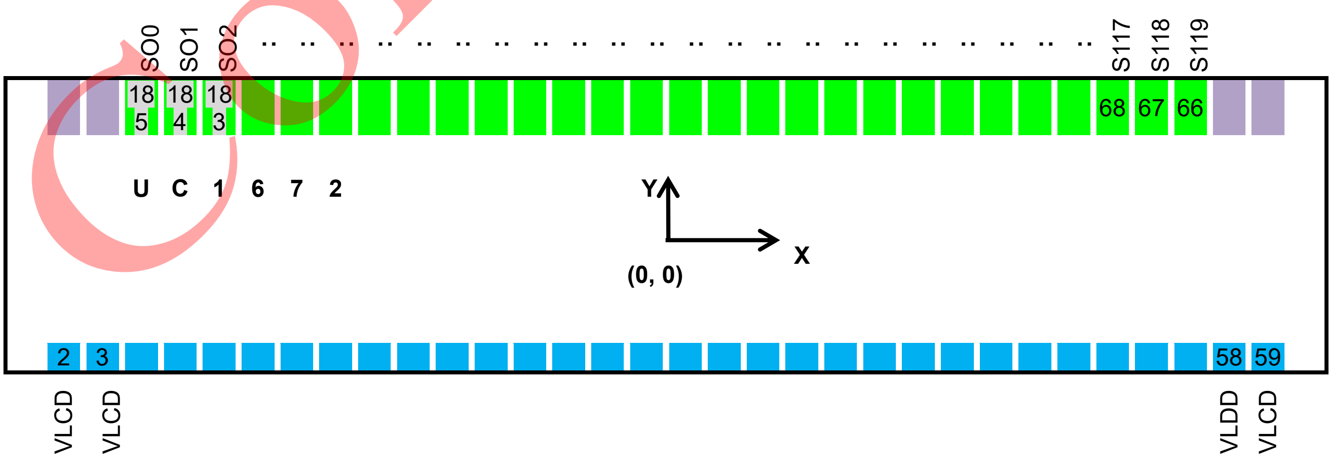
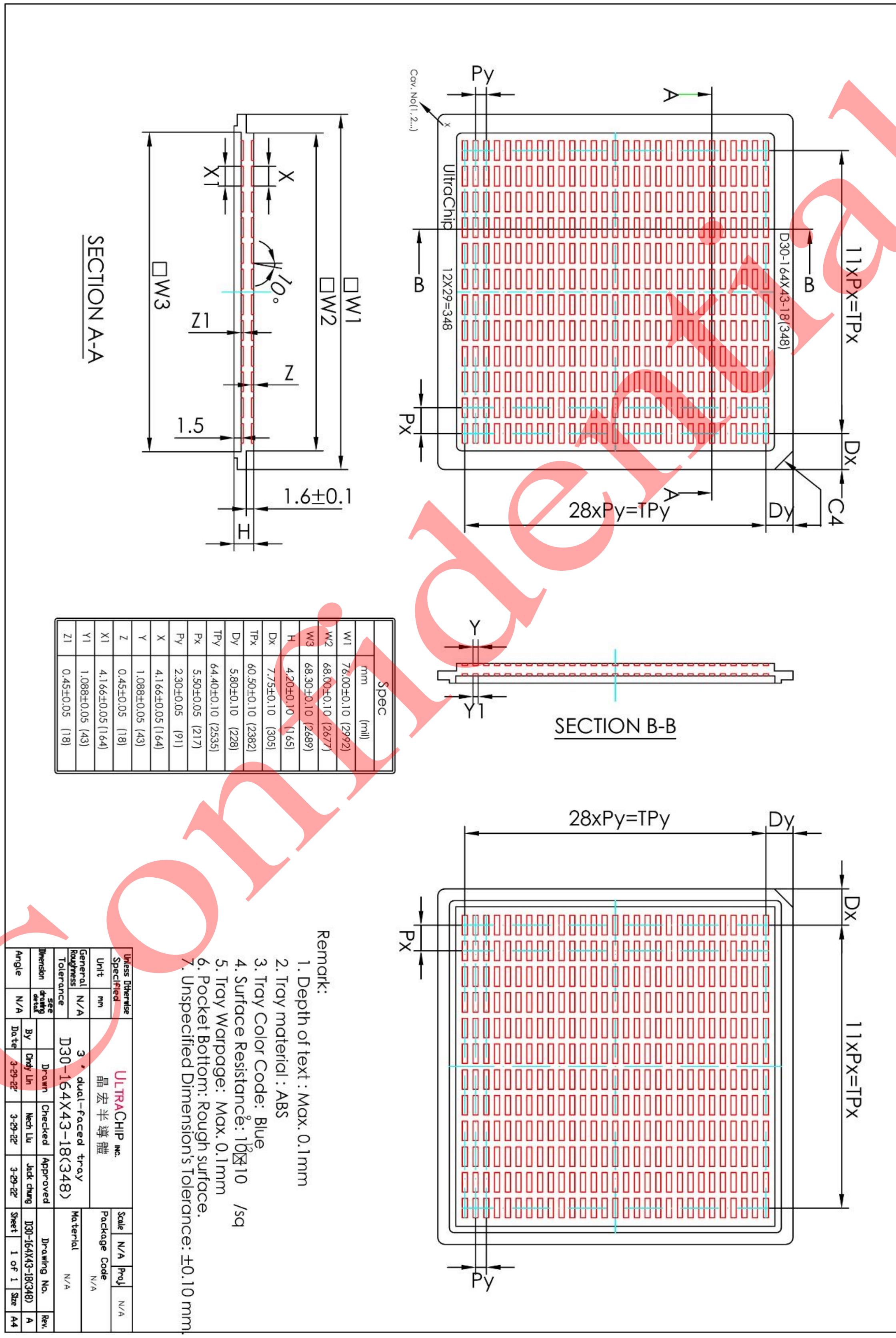
No	Name	X	Y	W	H
149	SEG<36>	-705	389	15	100
150	SEG<35>	-735	389	15	100
151	SEG<34>	-765	389	15	100
152	SEG<33>	-795	389	15	100
153	SEG<32>	-825	389	15	100
154	SEG<31>	-855	389	15	100
155	SEG<30>	-885	389	15	100
156	SEG<29>	-915	389	15	100
157	SEG<28>	-945	389	15	100
158	SEG<27>	-975	389	15	100
159	SEG<26>	-1005	389	15	100
160	SEG<25>	-1035	389	15	100
161	SEG<24>	-1065	389	15	100
162	SEG<23>	-1095	389	15	100
163	SEG<22>	-1125	389	15	100
164	SEG<21>	-1155	389	15	100
165	SEG<20>	-1185	389	15	100
166	SEG<19>	-1215	389	15	100
167	SEG<18>	-1245	389	15	100
168	SEG<17>	-1275	389	15	100
169	SEG<16>	-1305	389	15	100
170	SEG<15>	-1335	389	15	100

No	Name	X	Y	W	H
171	SEG<14>	-1365	389	15	100
172	SEG<13>	-1395	389	15	100
173	SEG<12>	-1425	389	15	100
174	SEG<11>	-1455	389	15	100
175	SEG<10>	-1485	389	15	100
176	SEG<9>	-1515	389	15	100
177	SEG<8>	-1545	389	15	100
178	SEG<7>	-1575	389	15	100
179	SEG<6>	-1605	389	15	100
180	SEG<5>	-1635	389	15	100
181	SEG<4>	-1665	389	15	100
182	SEG<3>	-1695	389	15	100
183	SEG<2>	-1725	389	15	100
184	SEG<1>	-1755	389	15	100
185	SEG<0>	-1785	389	15	100
186	BP_L<3>	-1845	389	15	100
187	BP_L<2>	-1875	389	15	100
188	BP_L<1>	-1905	389	15	100
189	BP_L<0>	-1935	389	15	100
190	NC<3>	-1965	389	15	100



TRAY INFORMATION



	Spec	
	mm	(mil)
W1	76.00±0.10	(2992)
W2	68.00±0.10	(2677)
W3	68.30±0.10	(2689)
H	4.20±0.10	(165)
Dx	7.75±0.10	(305)
TPx	60.50±0.10	(2382)
Dy	5.80±0.10	(228)
TPy	64.40±0.10	(2535)
Px	5.50±0.05	(217)
Py	2.30±0.05	(91)
X	4.166±0.05	(164)
Y	1.088±0.05	(43)
Z	0.45±0.05	(18)
X1	4.166±0.05	(164)
Y1	1.088±0.05	(43)
Z1	0.45±0.05	(18)

Revision History

Revision	Description	Date
0.6	(First release)	Jun. 13,2023